

# Separating $\Pi_{n+2}^1$ and $\Sigma_{n+2}^1$ Maximal Almost Disjoint Families

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## Abstract

We construct, for every  $n \geq 1$ , a forcing extension of  $M_n$  satisfying CH in which there is an infinite parameter-free  $\Pi_{n+2}^1$   $\omega$ -mad family and no infinite boldface  $\Sigma_{n+2}^1$  mad family.

## 1 Introduction

A family  $\mathcal{A} \subseteq [\omega]^\omega$  is *almost disjoint* if  $|a \cap b| < \omega$  whenever  $a, b \in \mathcal{A}$  are distinct. It is *mad* if for every  $x \in [\omega]^\omega$  there is  $a \in \mathcal{A}$  with  $|a \cap x| = \omega$ . An almost disjoint family is  $\omega$ -*mad* if, whenever  $\langle c_i : i < \omega \rangle$  is a sequence of infinite subsets of  $\omega$  and no  $c_i$  is covered modulo finite by a finite subfamily of  $\mathcal{A}$ , there is  $a \in \mathcal{A}$  such that  $|a \cap c_i| = \omega$  for every  $i$ . Every nonempty  $\omega$ -mad family is mad. All mad families in the statement below are required to be infinite.

At which projective level can a mad family first occur? At the first level, Mathias proved that there is no analytic mad family [11]. Miller subsequently proved that  $V = L$  implies the existence of a  $\Pi_1^1$  mad family [12]. Thus the existence of a coanalytic mad family is consistent, whereas the existence of an analytic mad family is ruled out in ZFC.

Friedman–Zdomskyy constructed projective  $\omega$ -mad families by combining stationary-set coding and almost disjoint coding [5]. Fischer–Friedman–Zdomskyy extended this approach to models with large continuum. They obtained a model with  $\mathfrak{b} = \mathfrak{c} = \omega_3$ , a  $\Delta_3^1$  wellorder of the reals, and a  $\Pi_2^1$   $\omega$ -mad family [4]. These results concern the coexistence of projective definitions with requirements on the continuum and on the maximality of the family.

At the second projective level, Törnquist proved that the existence of a  $\Sigma_2^1(a)$  mad family implies the existence of a  $\Pi_1^1(a)$  mad family [20]. In particular, a  $\Sigma_2^1$  mad family cannot be the first occurrence of a projective mad family. The theorem concerns the existence of another family; it does not assert that the original  $\Sigma_2^1(a)$  family is coanalytic.

The nonexistence results go beyond the analytic case. Törnquist proved that there are no mad families in Solovay’s model. He also showed that  $\omega_1^{L[a]} < \omega_1$  implies that there is no  $\Sigma_2^1(a)$  mad family [21]. Bakke Haga–Schrittesser–Törnquist developed an analysis using Mathias forcing relative to ideals and trees invariant under finite changes of a real. Among its consequences is that projective determinacy implies that there are no projective mad families [1]. Thus there are both constructions of projective mad families and hypotheses which rule out such families at several levels, or throughout the projective hierarchy.

For wellorders of the reals, the corresponding placement problem is settled by Steel’s analysis of the mice  $M_n$ . He showed that the reals of  $M_n$  are exactly the reals which are  $\Delta_{n+2}^1$  in a countable ordinal and, in particular, that  $M_n$  has a  $\Delta_{n+2}^1$  wellorder of the reals [16]. The mice with finitely many Woodin cardinals therefore calibrate the projective complexity of wellorders. For mad families, the question is whether one can prescribe the first projective level at which a family exists, rather than only obtain a family with a definition at that level.

We prove the following theorem.

**Theorem 1.1.** *Let  $n \geq 1$  and assume that  $M_n$  exists. There is a set-generic extension  $V^*$  of  $M_n$  satisfying CH such that:*

- (1) *there is an infinite parameter-free  $\Pi_{n+2}^1$ -definable  $\omega$ -mad family  $\mathcal{A}^* \subseteq [\omega]^\omega$ ;*
- (2) *there is no infinite boldface  $\Sigma_{n+2}^1$  mad family.*

For  $n = 1$  this gives a separation at the third projective level in a forcing extension of  $M_1$ . The same construction works over each  $M_n$ . The positive assertion requires no real parameter, while the negative assertion allows arbitrary real parameters. In particular, the latter excludes mad families in every lower projective pointclass.

## Outline of the proof

We work directly over  $M_n$  and use a countable-support iteration of length  $\omega_1$ . It alternates between two kinds of stages. At self-coding stages we add members to the family being constructed. At destruction stages we force that a given  $\Sigma_{n+2}^1$  almost disjoint family is not maximal. Bookkeeping treats all countable hitting requirements and all projective definitions, including their real parameters.

The self-coding stages modify the constructions of Friedman–Zdomskyy and Fischer–Friedman–Zdomskyy [5, 4]. A new real  $a$  is almost disjoint from the family constructed so far, meets a prescribed countable collection of positive sets, and codes itself into a block of stationary subsets of  $\omega_1$ . For each bit of  $a$ , the block has two stationary coordinates. An auxiliary forcing prepares a club code for each choice, but the actual iterand is the complete Boolean subforcing generated by  $a$ . Its extension is  $V[a]$ , not the whole auxiliary extension. The real  $a$  recovers the clubs at the coordinates it selects. Each club is localized independently, using David’s method [3]. An unselected coordinate can therefore be omitted from the auxiliary forcing. This gives relative properness on the corresponding Boolean cone and preserves the unselected stationary set.

To define the family, we use Steel’s results [15, 16, 17]. The lower part  $M_n|_{\omega_1}$  supplies the canonical wellorder, diamond sequence, stationary sets, and almost disjoint array. The sound projectum- $\omega$  initial segments of  $M_n$  compute the bounded parts of these objects. The resulting formula which asserts that a real got coded has complexity  $\Pi_{n+2}^1$  and defines the stage reals without parameters.

At destruction stages we use the relative Mathias forcing and Branch Lemma of Bakke Haga–Schrittesser–Törnquist [1]. The Woodin cardinals supply trees representing  $\Sigma_{n+2}^1$  definitions which remain correct through forcing extensions of the required size. The representations are obtained by the finite Martin–Steel homogeneity argument, complementation, and projection [10, 19, 18]. Once a parameter appears, the corresponding section is a fixed tree. Relative Mathias forcing adds a real almost disjoint from its full later projection, not merely from the family present at that stage.

Finally, the iteration is  $S_*$ -proper for a stationary reserve  $S_*$  and has a dense subset of size at most  $\omega_1$ . These facts give preservation of  $\omega_1$  and CH. The size bound also preserves mouse recognition and the projective tree representations. For each stationary coordinate, we prove relative properness below every condition forcing that it remains unselected. This preserves the coding pattern and excludes accidental self-codes. Bookkeeping then makes the family  $\omega$ -mad and destroys every candidate for a boldface  $\Sigma_{n+2}^1$  mad family.

The paper is organized as follows. In sections 2 and 3 we fix the coding conventions, prove mouse recognition, and define the canonical objects. Sections 4 and 5 contain the stationary coding sequence and the local decoder. The self-coding forcing is treated in section 6. In section 7 we give the relative Mathias forcing argument. Section 8 contains the iteration and its preservation proof. Sections 9 to 11 establish definability,  $\omega$ -maximality, and the nonexistence of boldface  $\Sigma_{n+2}^1$  mad families. We end with a list of open problems.

## 2 Coding conventions

The coding restraints start on a tail chosen after the positive sets are given. An arithmetic header in the mad real specifies its starting ordinal. The half-channel invariant supplies the padding needed to write this header.

The decoder uses a lower part of height  $\omega_1^N$ , not a projecting mouse with the same first uncountable ordinal as  $N$ .

**Lemma 2.1.** *Let  $P$  be externally sound and project to  $\omega$ . Suppose  $N$  is transitive,  $P \in N$ , and  $N$  supports set-structure satisfaction and the Skolem constructions for  $P$ . Then  $N$  regards  $P$  as countable. If  $\omega_1^P$  exists, it is less than  $\omega_1^N$ .*

*Proof.* Soundness and projection supply a finite degree and a finite parameter  $p$  whose Skolem hull of  $\omega \cup \{p\}$  is  $P$ . Since  $p \in N$ ,  $N$  can form the same Skolem map. Satisfaction for the fixed set structure and the assertion that the map has range  $P$  are absolute. Thus  $N$  contains an enumeration of  $P$ . It need not recognize  $p$  as standard or assert that  $P$  is sound.  $\square$

We therefore use a union of recognized segments and require its height to equal  $\omega_1^N$ . Recognition is evaluated externally, and the union is not required to project to  $\omega$ .

The tree representation used at destruction stages is obtained by Martin–Solovay complementation and projection. No homogeneity is claimed for the final projected tree.

## 3 The ground mouse and the canonical objects

Fix  $1 \leq n < \omega$ . Write  $M = M_n$  for Steel’s canonical mouse with  $n$  Woodin cardinals, and put

$$\kappa = \omega_1^M, \quad P_n = M \restriction \kappa.$$

Let  $\delta_0$  be its least Woodin cardinal. We use the construction, comparison, and correctness results of [15, 16, 17]. In particular,  $M \models \text{GCH}$ . Restrictions  $M \restriction \alpha$  are indexed by ordinal height; we also write  $\mathcal{J}_\alpha^{M_n}$  for these restrictions.

### 3.1 Countable initial segments

**Definition 3.1.** Let  $\text{It}_n(P, \omega)$  be Steel’s finite-level  $\Pi_n$  iterability predicate, with the even and odd definitions of [16, Definitions 1.4 and 1.6]. Fix its formula from [16, Lemmas 1.5 and 1.7]. Let  $\mathcal{I}_n$  consist of the real codes for well-founded unrelativized premice  $P$  such that

$$P \text{ is } \omega\text{-sound}, \quad \rho_\omega(P) = \omega, \quad P \text{ is } n\text{-small}, \quad \text{It}_n(P, \omega).$$

Here  $n$ -smallness means that, for each extender critical point  $\nu$  on the  $P$ -sequence,

$$\mathcal{J}_\nu^P \not\models \text{“there are } n \text{ Woodin cardinals.”}$$

Membership of a premouse in  $\mathcal{I}_n$  means membership of a code for it. For an ambient model  $W$ , write

$$\mathcal{C}_n^W = \{P_p : p \in \mathbb{R}^W \text{ and } W \models p \in \mathcal{I}_n\}$$

for the class of transitive structures so recognized.

**Lemma 3.2.** *The class  $\mathcal{I}_n$  is parameter-free  $\Pi_{n+1}^1$  and invariant under isomorphism of premouse codes.*

*Proof.* Steel's  $\Pi_n$  is the Levy hierarchy over HC; in real codes it becomes  $\Pi_{n+1}^1$  [16, Section 1]. His Lemmas 1.5 and 1.7 give this bound for the iteration predicate; the finite large-cardinal background supplies the determinacy used in the even case. The structural checks do not increase the bound, and well-foundedness is  $\Pi_1^1$ . The definition concerns the premouse, not its enumeration.  $\square$

**Lemma 3.3.** *Every member of  $\mathcal{C}_n^M$  is an initial segment of  $M$  of height below  $\kappa$ . These heights are unbounded in  $\kappa$ , and*

$$\bigcup \mathcal{C}_n^M = P_n.$$

*Proof.* Let  $U$  be a background for Steel's construction of  $M$ . The realizable sound projectum- $\omega$  segments in [16, Theorem 2.3] capture every real of  $M$ . They are countable in  $M$ , and their heights are unbounded in  $\kappa$ : a bound would put all reals of  $M$  in one countable level. See also [17, Corollaries 3.12–3.13] for the projectum- $\omega$  comparison and the appearance of new reals.

By [16, Corollary 4.4],  $M$  is  $\Sigma_n^{\text{HC}}$ -correct in  $U$ . Thus its  $\Pi_n^{\text{HC}}$  iterability assertions agree with those of  $U$ . Given  $P \in \mathcal{C}_n^M$ , choose a realizable segment  $Q$  as above with  $\text{Ord}^P < \text{Ord}^Q$ . Both are  $\omega$ -mice in the sense of [16, Definition 1.3]; they agree below  $\omega$ , which is a cutpoint of both. Lemma 2.2(3) of that paper gives  $P \trianglelefteq Q$  or  $Q \trianglelefteq P$ . The height inequality excludes the second alternative. Conversely, the cofinal realizable segments satisfy  $\text{It}_n$  by Lemma 2.2(2), and correctness transfers this to  $M$ . They belong to  $\mathcal{C}_n^M$ , proving the union assertion.  $\square$

The predicate contains no lower-part or recovery clause referring to  $M$ . Preservation of this recognition will follow from the tree representation in proposition 3.5.

### 3.2 Diamond and almost disjoint coding

Use the construction order  $<_{\text{can}}$  of  $M$ . Its restriction to an initial segment is that segment's construction order; extending the hierarchy inserts no elements before those already present.

**Lemma 3.4.** *There is a sequence*

$$\vec{D} = \langle D_\alpha : \alpha < \kappa \rangle$$

*in  $M$ , uniformly defined over  $P_n$  from  $<_{\text{can}}$ , which witnesses  $\diamond_\kappa$  in  $M$ .*

*Proof.* We use Jensen's least-pair recursion, as in [17, Theorem 5.4(1)]. Set  $D_\alpha = \emptyset$  at zero and successors. At a nonzero limit  $\alpha$ , choose the  $<_{\text{can}}$ -least pair  $(X, C)$  such that  $X \subseteq \alpha$ ,  $C$  is club in  $\alpha$ , and

$$\forall \xi \in C \quad X \cap \xi \neq D_\xi,$$

and put  $D_\alpha = X$ . Put  $D_\alpha = \emptyset$  if there is no such pair.

Suppose towards a contradiction that  $(X, C)$  is the least counterexample at  $\kappa$ . Choose an  $\omega$ -sound initial segment  $Q \triangleleft M$  containing  $\vec{D}, X, C, \kappa$  such that

$$\rho_\omega(Q) = \kappa = \omega_1^Q.$$

Such segments occur cofinally below  $M|\kappa^+$ : the appearance of a new subset of  $\kappa$  gives a level projecting to at most  $\kappa$ , and soundness excludes a smaller projectum for a level containing  $\kappa$ . New subsets of  $\kappa$  appear cofinally there, since each earlier level has size at most  $\kappa$ , whereas  $M \models 2^\kappa = \kappa^+$ . This is the projectum calculation used in [17, Sections 2 and 5].

In  $M$ , take a countable  $N \prec Q$  containing these parameters and the standard parameters of  $Q$  up to a degree witnessing  $\rho_\omega(Q) = \kappa$ . Put  $\delta = N \cap \kappa$ , and let  $\pi : \bar{Q} \rightarrow Q$  be the uncollapse map. Then

$$\text{crit}(\pi) = \delta = \rho_\omega(\bar{Q}) = \omega_1^{\bar{Q}}.$$

Apply [17, Theorem 5.1]. Its ultrapower alternative would give an extender  $E$  of length  $\delta$  and a degree-zero ultrapower of  $Q$ . If  $\nu = \text{crit}(E)$ , then  $\nu < \delta < \kappa$ , so a surjection  $g : \omega \twoheadrightarrow \nu$  belongs to  $Q$ . For the ultrapower map  $j$ , the function  $j(g)$  would have range  $j(\nu)$ , but  $j(g)(k) = j(g(k)) = g(k)$  for every  $k < \omega$ . This is impossible. Consequently  $\bar{Q} \triangleleft Q \triangleleft M$ .

The collapse sends  $\vec{D}$  to  $\vec{D} \upharpoonright \delta$  and  $(X, C)$  to  $(X \cap \delta, C \cap \delta)$ . By elementarity, the latter is the least counterexample at  $\delta$  in  $\bar{Q}$ . The counterexample test is bounded in its arguments. Moreover, any pair preceding this pair in  $<_{\text{can}}$  already belongs to  $\bar{Q}$ . Thus it is also the least counterexample at  $\delta$  in  $M$ , and the recursion gives  $D_\delta = X \cap \delta$ . Since  $C \in N$  is club in  $\kappa$ ,  $C \cap \delta$  is unbounded in  $\delta$  and  $\delta \in C$ . This contradicts the choice of  $(X, C)$ .

For each  $\alpha < \kappa$ , the choices through  $\alpha$  and their recursion history are hereditarily countable in  $M$ , and occur below  $\kappa$ . The construction order and the bounded witness test agree in every higher segment containing this history. Existence of such a history therefore defines  $D_\alpha$  uniformly over  $P_n$ .  $\square$

We also fix the array

$$\mathcal{B} = \{B_{\xi, m} : \xi < \kappa, m < \omega\} \subseteq [\omega]^\omega$$

obtained from the construction-order enumeration of  $2^\omega \cap M$  by the Jensen–Solovay recoding [8]. More precisely, let  $\langle r_\zeta : \zeta < \kappa \rangle$  be this enumeration, fix a recursive injection  $c : 2^{<\omega} \rightarrow \omega$ , and put

$$B_{0,0} = 2^\omega, \quad B_{\xi, m} = \{2c(r_{\omega\xi+m} \upharpoonright k) + 1 : k < \omega\} \quad ((\xi, m) \neq (0, 0)).$$

Distinct reals have only finitely many common initial segments, so this is an almost disjoint array. Write  $B_\xi = B_{\xi, 0}$  when only one channel is needed. Agreement of these computations between aligned segments is recorded in lemma 4.2.

### 3.3 Stable trees for projective definitions

**Proposition 3.5** (Stable projective trees from the finite homogeneity ladder). *Suppose that  $V$  has Woodin cardinals  $\delta_0 < \dots < \delta_{n-1}$ , where  $1 \leq n < \omega$ . Fix a cardinal  $\rho$  with  $\omega_1 < \rho < \delta_0$ . There is a tree  $T$  on  $\omega \times \lambda$ , for some ordinal  $\lambda$ , such that, for a fixed universal  $\Sigma_{n+2}^1$  formula  $U$ ,*

$$V[G] \models p[T] = \{v \in \omega^\omega : U(v)\}$$

*whenever  $G$  is generic over  $V$  for a forcing having in  $V$  a dense subset of cardinality less than  $\rho$ . The input  $v$  codes the formula index, real parameter, and argument. Consequently, taking a section after a parameter first appears produces a fixed tree representing that definition in all later extensions which, viewed over  $V$ , are obtained by forcings having a dense subset of size less than  $\rho$ .*

*Proof.* Replacing the forcing by a dense suborder reduces the assertion to forcings whose underlying set has size less than  $\rho$ .

*The case  $n = 1$ .* Choose a measurable  $\mu$  with  $\rho < \mu < \delta_0$ ; this is possible because a Woodin cardinal is a limit of measurable cardinals. Write the universal  $\Sigma_3^1$  formula as

$$U(v) \iff \exists y \neg \exists z R(v, y, z), \quad R \in \Pi_1^1.$$

Martin’s homogeneous representation gives a  $\mu$ -homogeneous Shoenfield tree  $H$  for  $R$ . Projecting the real  $z$  gives a  $\mu$ -weakly homogeneous tree  $S$  for  $\exists z R(v, y, z)$ . Take its Martin–Solovay complement  $D$ , then project the real  $y$  to obtain  $T$ . The coanalytic ranking-tree interpretation and the complementing property remain correct in all forcings of size less than  $\mu$ . Thus in those extensions

$$p[T] = \{v : \exists y (v, y) \in p[D]\} = \{v : \exists y \neg \exists z R(v, y, z)\}.$$

This proves the desired assertion, in fact with the larger bound  $\mu$ . This is exactly the measurable-cardinal construction recalled by Steel in *Games and Scales*, Section 2.1.

*The case  $n \geq 2$ .* Choose a measurable  $\mu$  with  $\delta_{n-2} < \mu < \delta_{n-1}$ , again using that the top Woodin is a limit of measurables. Martin's homogeneous representation of coanalytic sets gives, uniformly for each  $\Pi_1^1$  formula, a  $\mu$ -homogeneous Shoenfield tree. The representation is correct in generic extensions by forcings of size less than  $\mu$ .

We use the following two standard operations, keeping the represented formula as part of the induction invariant. First, real projection of a  $\kappa$ -homogeneous tree is  $\kappa$ -weakly homogeneous: the projected real witness is absorbed into the tree's ordinal coordinate. Second, the Martin–Solovay construction produces a  $\kappa$ -absolute complement of a  $\kappa$ -weakly homogeneous tree. If  $\delta$  is Woodin and the weak homogeneity system is  $\delta^+$ -complete, the Martin–Steel theorem additionally makes a sufficiently tall Martin–Solovay complement  $\kappa$ -homogeneous for any prescribed  $\kappa < \delta$ .

Starting with  $\Pi_1^1$ , apply this last complement-of-projection operation  $n - 1$  times, using in descending order  $\delta_{n-2}, \dots, \delta_0$ . At each nonfinal step choose the new completeness above the next Woodin; at the final step choose completeness  $\rho$ . This gives a  $\rho$ -homogeneous tree for every  $\Pi_n^1$  formula, uniformly including its real parameters among the input coordinates.

These are representations by the actual formulas in small generic extensions, not merely representations of their ground-model truth sets. To verify this, induct on the displayed syntactic construction. The base ranking tree represents the coanalytic formula in the extension. Real projection implements the existential quantifier in the extension. The Martin–Solovay complement implements negation there, because the pair remains complementary. Homogeneity systems lift through the small forcing, so the invariant continues at the next step. This is the finite version of the formula-tree induction in Steel's proof of the generic absoluteness theorem.

Now put the universal formula in the form

$$U(v) \iff \exists y \neg \exists z R(v, y, z), \quad R \in \Pi_n^1.$$

Let  $H$  be the  $\rho$ -homogeneous tree just constructed for  $R$ . Project the  $z$  coordinate to obtain a  $\rho$ -weakly homogeneous tree  $S$  for  $\exists z R(v, y, z)$ . Let  $D$  be its Martin–Solovay complement. Finally project the  $y$  coordinate of  $D$ , obtaining  $T$ . In every extension of the stipulated size,

$$p[T] = \{v : \exists y (v, y) \in p[D]\} = \{v : \exists y \neg \exists z R(v, y, z)\}.$$

The last real projection is used only as a tree projection. No homogeneity or absolute-complement assertion is made for  $T$ .  $\square$

*Remark 3.6* (References for the two operations). J. R. Steel, *The derived model theorem*, online version [19]: Theorem 1.20 and Corollary 1.21 (printed pp. 8–9) give the Martin–Solovay absolute complement and lifting; Theorem 2.1 and Corollary 2.3 (printed pp. 9–10) give Martin–Steel propagation. The proof of Theorem 2.6 (printed pp. 10–11) explicitly carries the formula-tree induction through small generic extensions. Steel, *Games and Scales*, [18], printed p. 3 recalls the homogeneous coanalytic base, and Section 2.1, printed p. 4, explicitly states the measurable-cardinal version for  $\Sigma_3^1$  formulas and all sufficiently small forcing extensions.

*Remark 3.7* (Application to the iteration). Inside  $M_n$  take  $\rho = (\kappa^+)^{M_n}$ , below its least Woodin cardinal. By theorem 8.5, every initial segment and the completed iteration have a dense subset of size at most  $\kappa$  in  $M_n$ . The proposition therefore applies to each of these forcings. No bound on the size of the Boolean completion is needed.

Fix the tree given by the proposition. For a parameter  $r$  first appearing in  $V_\alpha$ , let  $T(e, r)$  be its section. This is the same tree throughout the rest of the iteration, and its projection represents the formula in each later model. The fixed-tree argument of [1, Lemma 2.3 and Main Proposition 3.5] uses neither homogeneity nor a complementing tree.

**Lemma 3.8.** *Under the forcing size bound of proposition 3.5,  $T = T(e, r)$  continues to represent the formula along the relevant tail. Independently of that requirement, for a fixed tree  $T$  the assertions that  $p[T] \subseteq [\omega]^\omega$  and that  $p[T]$  is almost disjoint are absolute to forcing extensions. For a fixed old real  $y$ , membership  $y \in p[T]$  is also absolute.*

*Proof.* The first assertion follows from the preceding proposition. A violation of almost disjointness or of the domain restriction is witnessed by a branch through a fixed auxiliary tree. For an old  $y$ , use its fixed section  $T_y$ . Well-foundedness of these set trees is absolute, since a well-founded tree has an ordinal rank function. See [1, Lemma 2.3].  $\square$

### 3.4 Recognition in forcing extensions

**Theorem 3.9** (Recognition and preservation). *Suppose  $\mathbb{P} \in M$ ,  $M \models \text{dens}(\mathbb{P}) < \delta_0$ , and  $\mathbb{P}$  preserves  $\kappa$ . If  $G$  is  $M$ -generic, then*

$$\mathcal{C}_n^{M[G]} = \mathcal{C}_n^M, \quad \bigcup \mathcal{C}_n^{M[G]} = P_n.$$

*Thus the recognized structures and their cofinal set of heights are unchanged. New real codes for these structures are allowed.*

*Proof.* Choose in  $M$  a cardinal  $\rho$  with  $\max\{\kappa, \text{dens}(\mathbb{P})\} < \rho < \delta_0$ . By proposition 3.5, a fixed tree represents the universal  $\Sigma_{n+2}^1$  formula in both models. For an old real input, its section either has an old branch or has an old ordinal rank function. Hence  $M$  and  $M[G]$  agree on  $\Sigma_{n+2}^1$  statements with old real parameters, and on statements at lower levels.

For well-founded premouse codes, let  $\text{Comp}(p, q)$  say that  $P_p \trianglelefteq P_q$  or  $P_q \trianglelefteq P_p$ . An initial-segment isomorphism is witnessed by a real, so this relation has an analytic presentation. For an old recognized code  $q$ , the statement

$$\exists p (p \in \mathcal{I}_n \wedge \neg \text{Comp}(p, q)) \tag{*}$$

is  $\Sigma_{n+2}^1$ . Ground recognition makes it false in  $M$ ; by absoluteness it is false in  $M[G]$ .

Let  $M[G] \models p \in \mathcal{I}_n$ . Since  $\kappa$  is preserved,  $\text{Ord}^{P_p} < \kappa$ . Choose an old recognized segment  $Q$  of larger height and an old code  $q$  for it. The failure of (\*) gives  $P_p \trianglelefteq Q$ . Thus  $P_p$  is an old structure. Choose an old code  $p_0$  for it. Isomorphism invariance and old-parameter absoluteness give  $M \models p_0 \in \mathcal{I}_n$ . Conversely, every old recognized code remains accepted. This proves both assertions.  $\square$

The height bound is used to choose a taller old segment. The theorem makes no recognition assertion in outer models which collapse  $\omega_1^M$ .

## 4 The stationary coding sequence

### 4.1 The sequence

For  $i < \omega_1$ , let  $A_i = \{i\}$  and let  $A_* = \emptyset$ . Define

$$S_i = \{\alpha \in \text{Lim}(\omega_1) : D_\alpha = A_i \cap \alpha\}$$

and

$$S_* = \{\alpha \in \text{Lim}(\omega_1) : D_\alpha = \emptyset\}.$$

The set  $S_*$  is reserved and is never killed.

**Lemma 4.1.** *The family*

$$\{S_*\} \cup \{S_i : i < \omega_1\}$$

*has the following properties in  $M_n$ .*

- (1) Every member is stationary and co-stationary.
- (2) If  $i \neq j$ , then  $S_i \cap S_j$  is bounded.
- (3) For every  $i$ ,  $S_i \cap S_*$  is bounded.
- (4) The sequence is uniformly definable over  $P_n$ .

*Proof.* Stationarity follows from diamond applied to  $A_i$  and to  $A_*$ . If  $i \neq j$  and  $\alpha > \max\{i, j\}$ , then  $A_i \cap \alpha \neq A_j \cap \alpha$ , so  $\alpha$  cannot belong to both  $S_i$  and  $S_j$ . The same argument applies to  $S_i$  and  $S_*$ . Since each  $S_i$  has a stationary set almost disjoint from it, it is co-stationary. Uniform definability follows from lemma 3.4.  $\square$

For  $\eta < \omega_1$ ,  $k < \omega$ , and  $\varepsilon < 2$ , put

$$\iota(\eta, k, \varepsilon) = \omega\eta + 2k + \varepsilon.$$

The  $\eta$ -th coding block is

$$\mathcal{S}^\eta = \langle S_{\iota(\eta, k, 0)}, S_{\iota(\eta, k, 1)} : k < \omega \rangle.$$

## 4.2 Agreement of canonical computations

A computation in a segment means a completed history of the recursions above. If a segment does not yet contain a least-pair witness at a limit step, that computation is not completed; it does not supply a final empty-set assignment.

**Lemma 4.2.** *Aligned initial segments of  $M$  agree on completed computations of the construction order, its real enumeration,  $\vec{D}$ ,  $\mathcal{B}$ , and  $\langle S_i : i < \kappa \rangle$ , including  $S_*$ . For each  $\alpha < \kappa$  there is a recognized segment in which the data through  $\alpha$  and their computation histories occur. Every higher recognized segment gives the same values.*

*Proof.* The construction order restricts coherently. Thus completed initial parts of the real enumeration, and their fixed almost disjoint recodings, agree.

Given the previous diamond values, the test on a candidate  $(X, C)$  is bounded in  $\alpha, X, C$ , and those values. It is absolute between transitive segments containing them. Every candidate preceding a chosen witness is already in the smaller segment. Induction therefore preserves each least choice. A witness exists at every nonzero countable limit  $\alpha$ : choose a strictly increasing cofinal sequence  $\langle \gamma_j : j < \omega \rangle$  with  $\gamma_j > j$ , and put

$$C = \{\gamma_j : j < \omega\}, \quad X = \{j < \omega : j \notin D_{\gamma_j}\}.$$

Then  $C$  is club in  $\alpha$ , and  $X \cap \gamma_j \neq D_{\gamma_j}$  at coordinate  $j$ . Hence these limit computations are eventually completed. The definitions of  $S_i$  and  $S_*$  now give pointwise agreement; no local assertion of stationarity is needed.

Each bounded collection of outputs and histories is hereditarily countable in  $M$  and occurs below  $\kappa$ . Ground recognition gives a higher recognized segment containing it. The same agreement holds in the extensions of theorem 3.9, since their recognized structures are unchanged.  $\square$

Let  $H_n = \{\beta < \kappa : M \upharpoonright \beta \in \mathcal{C}_n^M\}$ , and choose in  $M$  a function  $f : \kappa \rightarrow H_n$  such that  $f(\alpha) > \alpha$  and  $M \upharpoonright f(\alpha)$  contains the computations through  $\alpha$  from the lemma. This function will be used only in the nonvacuity proof, not as a parameter in the decoder.



### 4.3 Club shooting

For  $i < \omega_1$ , let  $\mathbb{C}_i$  be the forcing whose conditions are countable closed bounded subsets of  $\omega_1 \setminus S_i$ , ordered by end-extension. The generic union is a club disjoint from  $S_i$ . This is the standard club-shooting forcing of Baumgartner–Harrington–Kleinberg [2].

We use relative properness.

**Definition 4.3.** Let  $R \subseteq \omega_1$  be stationary. A forcing  $\mathbb{P}$  is *R-proper* if, for every sufficiently large regular  $\theta$ , there is a club  $\mathcal{C} \subseteq [H_\theta]^\omega$  such that whenever  $M \in \mathcal{C}$ ,  $\mathbb{P} \in M$ , and  $M \cap \omega_1 \in R$ , every condition in  $\mathbb{P} \cap M$  has an  $(M, \mathbb{P})$ -generic extension.

**Lemma 4.4.** Let  $R \subseteq \omega_1$  be stationary. An *R-proper forcing preserves  $\omega_1$  and the stationarity of  $R$ . A countable-support iteration of forcings which are forced to be R-proper is R-proper. These assertions also apply to restrictions below conditions.*

*Proof.* Let  $p \in \mathbb{P}$ , and choose a countable  $N \prec H_\theta$  in the club witnessing *R*-properness, with  $p, \mathbb{P} \in N$  and  $\delta = N \cap \omega_1 \in R$ . An  $(N, \mathbb{P})$ -generic  $q \leq p$  forces  $N[G] \cap \omega_1 = \delta$ . Applying this to a name in  $N$  for a countable sequence of countable ordinals proves preservation of  $\omega_1$ . If  $\dot{C} \in N$  is a name for a club, then  $q$  forces  $\dot{C} \cap \delta$  to be unbounded in  $\delta$ . Hence  $q \Vdash \delta \in \dot{C} \cap R$ .

For the iteration assertion, use the countable-support iteration theorem from [14, Chapter III], with the test models restricted to heights in  $R$ . At an intermediate master condition,  $N[G_\beta] \cap \omega_1 = N \cap \omega_1 \in R$ , so the next relative properness assumption applies at the same height. Include names for the witnessing clubs among the parameters. At a limit, the proof constructs master prefixes along a sequence cofinal in  $N \cap \lambda$ , leaving each constructed prefix fixed. Their union has countable support and meets the dense sets in  $N$ . The proof is unchanged when the iteration is restricted along a condition. This use of relative properness in stationary coding also appears in [6].  $\square$

**Lemma 4.5.** Let  $R \subseteq \omega_1$  be stationary and suppose  $R \cap S_i \in \text{NS}_{\omega_1}$ . Then  $\mathbb{C}_i$  is *R-proper* and  $\omega$ -distributive. It preserves  $R$ . In particular,  $\mathbb{C}_i$  preserves  $S_*$  and every  $S_j$  with  $j \neq i$ .

*Proof.* Choose a club  $E$  disjoint from  $R \cap S_i$ , and take a countable  $N \prec H_\theta$  containing  $\mathbb{C}_i, R, E$  with  $\delta = N \cap \omega_1 \in R$ . Then  $\delta \in E$  and hence  $\delta \notin S_i$ .

Fix  $p \in \mathbb{C}_i \cap N$ . Enumerate the dense open subsets of  $\mathbb{C}_i$  in  $N$  as  $\langle U_k : k < \omega \rangle$ , and choose  $\langle \delta_k : k < \omega \rangle$  increasing and cofinal in  $\delta$ . Starting with  $p_0 = p$ , choose

$$p_{k+1} \in U_k \cap N, \quad p_{k+1} \leq p_k, \quad \max(p_{k+1}) > \delta_k.$$

This is possible by elementarity and the unboundedness of  $\omega_1 \setminus S_i$ . The set

$$q = \bigcup_{k < \omega} p_k \cup \{\delta\}$$

is closed and disjoint from  $S_i$ , and end-extends every  $p_k$ . Thus  $q$  is an  $(N, \mathbb{C}_i)$ -generic extension of  $p$ . This proves *R*-properness, and lemma 4.4 gives preservation of  $R$ .

Given any countable family of dense open sets and any condition  $p$ , choose such an  $N$  containing the family and  $p$ . The same construction gives a condition in all of those sets. Hence  $\mathbb{C}_i$  is  $\omega$ -distributive. Finally apply the result with  $R = S_*$  or  $R = S_j$ ,  $j \neq i$ , using lemma 4.1.  $\square$

**Lemma 4.6.** Let  $I \subseteq \omega_1$  be countable and let  $R$  be stationary, with  $R \cap S_i \in \text{NS}_{\omega_1}$  for every  $i \in I$ . Then

$$\mathbb{D}_I = \prod_{i \in I} \mathbb{C}_i$$

with countable support is *R-proper* and  $\omega$ -distributive. Under CH it has size at most  $\omega_1$ .

*Proof.* The empty product is trivial, so assume  $I \neq \emptyset$ . Choose a club  $E$  disjoint from  $R \cap \bigcup_{i \in I} S_i$ . Take a countable  $N \prec H_\theta$  containing the parameters and a condition  $p$ , with  $\delta = N \cap \omega_1 \in R$ . Then  $\delta \notin S_i$  for every  $i \in I$ .

Enumerate  $I$  and the dense open sets belonging to  $N$ , and choose  $\delta_k \nearrow \delta$ . Construct  $p_{k+1} \leq p_k$  in  $N$ , starting with  $p_0 = p$ , so that  $p_{k+1}$  belongs to the  $k$ -th dense set and its first  $k+1$  club coordinates have maxima above  $\delta_k$ . Put

$$q(i) = \bigcup_{k < \omega} p_k(i) \cup \{\delta\} \quad (i \in I),$$

where an unused coordinate has the empty condition. Each  $q(i)$  is closed, avoids  $S_i$ , and end-extends every  $p_k(i)$ . Thus  $q \leq p$  is  $N$ -generic. Including any prescribed countable family of dense open sets among the parameters proves  $\omega$ -distributivity as in the preceding lemma. Under CH, each factor has size at most  $\omega_1$  and  $(\omega_1)^{\aleph_0} = \omega_1$ , proving the size bound.  $\square$

**Lemma 4.7.** *Let  $x \in 2^\omega$  and let  $\eta < \omega_1$ . Let  $\mathbb{C}(\eta, x)$  be the countable-support iteration of length  $\omega$  which, at coordinate  $k$ , uses*

$$\mathbb{C}_{\iota(\eta, k, x(k))}.$$

*Then:*

- (1)  $\mathbb{C}(\eta, x)$  is  $S_*$ -proper and adds no reals;
- (2) it adds clubs  $C_k \subseteq \omega_1 \setminus S_{\iota(\eta, k, x(k))}$ ;
- (3) it preserves every  $S_j$  not selected by the displayed rule;
- (4) in its extension, the  $\eta$ -th block has the exact pattern

$$S_{\iota(\eta, k, \varepsilon)} \text{ is nonstationary} \iff \varepsilon = x(k).$$

*Proof.* Write  $i_k = \iota(\eta, k, x(k))$  and  $\mathbb{D} = \prod_{k < \omega} \mathbb{C}_{i_k}$ . By lemma 4.6, this product is  $S_*$ -proper and  $\omega$ -distributive. We show that its constant-coordinate presentation is dense in  $\mathbb{C}(\eta, x)$ .

Each finite iteration adds no countable sequences of ordinals, by induction using lemma 4.5. Thus its next club forcing has exactly its ground-model conditions, and each finite initial segment has a dense presentation by tuples of such conditions.

Fix  $p \in \mathbb{C}(\eta, x)$ . Choose a countable  $N \prec H_\theta$  containing  $p$  and the iteration, with  $\delta = N \cap \omega_1 \in S_*$  outside all the  $S_{i_k}$ . Let  $\delta_l \nearrow \delta$ . Construct  $p_{l+1} \leq p_l$  in  $N$ , starting with  $p_0 = p$ , such that the first  $l+1$  coordinates of  $p_{l+1}$  are decided as ground-model club conditions, each with maximum above  $\delta_l$ . Strengthen the finite prefix and retain the tail to obtain each  $p_{l+1}$ . For  $l > k$ , write  $c_{l,k}$  for the decided  $k$ -th coordinate and put

$$q(k) = \bigcup_{l > k} c_{l,k} \cup \{\delta\}.$$

These are ground-model conditions, since  $\delta \notin S_{i_k}$ . An induction on  $k$  gives  $q \restriction k \leq p_l \restriction k$  for every  $l$ : at coordinate  $k$ , use a later  $p_r \leq p_l$  whose  $k$ -th coordinate has been decided, and note that  $q(k)$  end-extends it. Hence  $q \leq p$ . The order on constant tuples is the product order, so the product is dense in the iteration.

The first assertion now follows from lemma 4.6. The union at coordinate  $k$  is club and disjoint from  $S_{i_k}$ . For an unselected  $j$ , apply the same product lemma with  $R = S_j$ ; its intersections with the countably many selected sets are nonstationary. Thus every unselected coordinate remains stationary. This proves the remaining assertions.  $\square$

## 5 The local decoder

Work in an extension covered by theorem 3.9; thus  $\omega_1 = \kappa$ . We use  $\mathcal{I}_n$  for projecting segments and  $\mathcal{J}_n$  for their limit closure. The computation lemma supplies the arrays used below.

### 5.1 Limit unions

**Definition 5.1.** Let  $\mathcal{J}_n$  consist of the real codes for nonempty well-founded premisses  $P$  such that, writing  $\delta = \text{Ord}^P$ ,

$$\forall \alpha < \delta \exists \beta (\alpha < \beta < \delta \wedge P| \beta \in \mathcal{I}_n).$$

An  $n$ -decoding pair for  $a$  is a pair  $(N, P)$  such that  $N$  is countable and transitive, satisfies the decoding fragment,  $a, P \in N$ , and

$$P \in \mathcal{J}_n \text{ externally,} \quad \text{Ord}^P = \omega_1^N.$$

The fragment includes set-structure satisfaction, Separation, the instances of Replacement used in decoding, and wellorder collapse.

The equality concerns the height of  $P$ , not  $\omega_1^P$ . Neither projection to  $\omega$  for  $P$  nor internal recognition by  $N$  is required. The structure  $P$  belongs to  $N$ ; its real code need not belong to  $N$ . Cofinal certification is external and does not require an  $\omega$ -sequence in  $N$  cofinal in  $\omega_1^N$ .

**Lemma 5.2.** *The class  $\mathcal{J}_n$  is parameter-free  $\Pi_{n+1}^1$ . Its members are exactly  $M|\delta$  for nonzero limit points  $\delta < \kappa$  of  $H_n$ .*

*Proof.* In a code for  $P$ , the quantifiers over  $\alpha, \beta$  are number quantifiers and the specified initial-segment codes are uniform. Thus the complexity follows from lemma 3.2. By theorem 3.9, the certified segments align, and

$$P = \bigcup \{P| \beta : \beta < \delta, P| \beta \in \mathcal{I}_n\} = M|\delta.$$

The converse follows by taking the union at a limit point of  $H_n$ . □

### 5.2 Decoding data

**Definition 5.3.** A code for  $(P, \vec{B}^P, \vec{S}^P)$  belongs to  $\mathcal{K}_n$  if  $P$  is a nonempty well-founded premiss of height  $\delta$ ,

$$\vec{B}^P = \langle B_{\xi, m}^P : \xi < \delta, m < \omega \rangle, \quad \vec{S}^P = \langle S_i^P : i < \delta \rangle,$$

where  $B_{\xi, m}^P \subseteq \omega$  and  $S_i^P \subseteq \delta$ , and for each  $\alpha < \delta$  there is  $\beta$  with  $\alpha < \beta < \delta$  such that  $P| \beta \in \mathcal{I}_n$  and a completed canonical computation in  $P| \beta$  agrees with the displayed data through  $\alpha$ . This includes every  $B_{\xi, m}^P$  for  $\xi < \alpha, m < \omega$ , and each membership statement  $\zeta \in S_i^P$  for  $i, \zeta < \alpha$ . Computation histories are elements of  $P| \beta$ , not separately quantified real codes.

A decoding tuple for  $a$  is  $(N, P, \vec{B}^P, \vec{S}^P)$  such that  $(N, P)$  is an  $n$ -decoding pair for  $a$ , the arrays belong to  $N$ , and  $(P, \vec{B}^P, \vec{S}^P) \in \mathcal{K}_n$  externally.

This certification records the arrays computed relative to the union  $P$ . It imposes no iterability condition on  $P$  itself.

**Lemma 5.4.** *The class  $\mathcal{K}_n$  is parameter-free  $\Pi_{n+1}^1$ . Every tuple it recognizes has*

$$P = M|\delta, \quad B_{\xi, m}^P = B_{\xi, m} \quad (\xi < \delta, m < \omega), \quad S_i^P = S_i \cap \delta \quad (i < \delta).$$

*For every real  $a$ , decoding tuples exist at club many heights  $\delta < \kappa$ .*

*Proof.* The segment and history quantifiers in the definition are number quantifiers in a code for  $P$ . Checking a completed computation in the coded structure is arithmetic. Together with lemma 3.2, this gives  $\Pi_{n+1}^1$ . The definition implies  $P \in \mathcal{J}_n$ , so lemma 5.2 identifies  $P$ ; agreement of the arrays follows from lemma 4.2.

For nonvacuity, fix in  $M$  the level map  $\alpha \mapsto M|\alpha$  and enumerations  $e_\alpha : \omega \twoheadrightarrow M|\alpha$  for  $0 < \alpha < \kappa$ . Choose a countable  $X \prec H_\theta$  containing  $a, P_n, \kappa$ , these maps, the arrays, and the function  $f$  from lemma 4.2. Put  $\delta = X \cap \kappa$ . Then

$$X \cap P_n = M|\delta.$$

Indeed, the level of an element of  $X \cap P_n$  belongs to  $X \cap \kappa$ . Conversely, for  $0 < \alpha < \delta$ , the enumeration  $e_\alpha \in X$  gives  $M|\alpha \subseteq X$ .

The collapse sends  $P_n$  to  $M|\delta$ ,  $\kappa$  to  $\omega_1^N = \delta$ , and the arrays to the displayed restrictions. Closure under  $f$  supplies cofinally many computation certificates below  $\delta$ . Hence the collapse is a decoding tuple. A continuous chain of such hulls gives club many heights.  $\square$

### 5.3 An arithmetic code for the starting ordinal

Normalize the array by taking its reserved member to be  $B_{0,0} = 2\omega$  and placing all other members on the odd integers. For  $a \cap 2\omega$  infinite, enumerate it increasingly and define

$$z_a(n) = 1 \iff (a \cap 2\omega)(2n) \in 4\omega.$$

This is an arithmetic operation on  $a$ . Use a fixed recursive convention for interpreting  $z_a$  as a binary relation on  $\omega$ . The header condition says that this relation is a wellorder of type  $\gamma_a \geq \omega$ . It is  $\Pi_1^1$ , and  $\gamma_a$  is uniquely determined by  $a$ .

If  $N$  is any decoding model containing such an  $a$ , then  $z_a \in N$ . Wellorder collapse in  $N$  computes the actual order type  $\gamma_a$ , and therefore

$$\gamma_a < \omega_1^N.$$

The forcing can prescribe a chosen header by requiring the even-indexed points of  $a \cap 2\omega$  to have the corresponding enumeration parity. Odd-indexed points provide padding. The two-half reserve invariant in the stage forcing is the hypothesis that makes these requirements extendible; the decoder itself needs only the displayed arithmetic definition.

### 5.4 Individual column codes and their markers

For an ordinal  $\eta < \omega_1$ , an integer  $m \geq 1$ , and a club  $C \subseteq \omega_1$ , define

$$\text{Pack}(\eta, m, C) = \{3\xi : \xi \in C\} \cup \{3\xi + 1 : \xi < \eta\} \cup \{3\ell + 2 : \ell < m\}.$$

Thus the three channels give the club, the initial segment  $\eta$ , and the finite initial segment  $m$ . The ordinal operations commute with restriction to  $\delta = \omega_1^N$  in any transitive decoding model: every ordinal occurring in the operations on arguments below  $\delta$  is countable in  $N$  and remains below  $\delta$ .

Fix  $\gamma > \omega\eta + \omega$ . Define  $d_0$  to be the least point of  $\text{Acc}(C)$  above  $\gamma$ , let  $d_{\alpha+1}$  be the least point of  $\text{Acc}(C)$  at least  $d_\alpha + \omega$ , and take suprema at limit indices. Put  $D = \{d_\alpha : \alpha < \omega_1\}$ . This is a club contained in  $\text{Acc}(C)$ , its enumeration is continuous, and adjacent points have the required gap. Fix in the ground model a wellorder code for each countable ordinal; use these fixed codes for the markers below. Thus the entire transform is a specified function of  $C, \eta, m, \gamma$ , and fixed ground parameters. Write  $\text{Odd}(Z) = \{\xi : 2\xi + 1 \in Z\}$  and  $\text{Even}(Z) = \{\xi : 2\xi \in Z\}$  for ordinal channels. For  $X = \text{Pack}(\eta, m, C)$  define  $Z = \text{Loc}_D(X)$  by the usual David arrangement: its

odd channel is  $X$ ; its even channel on the first  $\omega$  coordinates codes  $d_0$ ; on  $[d_\alpha, d_\alpha + \omega)$  it codes  $d_{\alpha+1}$ ; and the remaining even-channel positions in each gap are zero. Finally put

$$\text{TailLoc}_\gamma(X) = \{\gamma + \xi : \xi \in \text{Loc}_D(X)\} \subseteq [\gamma, \omega_1).$$

Only the single club  $C$ , the fixed ordinals, and its own marker club enter this transformation. In particular, a column does not depend on the other club coordinates or on the eventual real selecting it.

**Lemma 5.5.** *Suppose  $N$  is a transitive decoding model,  $\delta = \omega_1^N > \gamma$ , and  $Y \cap [\gamma, \delta) \in N$ , where  $Y = \text{TailLoc}_\gamma(\text{Pack}(\eta, m, C))$ . Then  $\delta \in D$ , and  $N$  decodes exactly  $\eta, m, C \cap \delta$ . In particular it sees the decoded club as closed and unbounded in  $\delta$ .*

*Proof.* The shift  $\xi \mapsto \gamma + \xi$  is a bijection from the ordinals onto the tail starting at  $\gamma$ , and it and its inverse commute with restriction to  $\delta$ . Thus  $N$  obtains  $\text{Loc}_D(X) \cap \delta$ . If  $\delta < d_0$ , its first marker would code a countable wellorder of type at least  $\delta$  inside  $N$ , a contradiction. If  $d_\alpha < \delta < d_{\alpha+1}$ , then  $d_\alpha + \omega < \delta$  because  $\delta$  is an uncountable cardinal of  $N$ . The marker on that interval is again a real in  $N$  coding an ordinal at least  $\delta$ , a contradiction. Hence  $\delta \in D$ . The odd channel gives  $X \cap \delta$ , and its three channels give exactly the stated objects. Since  $\delta \in \text{Acc}(C)$ , the set  $C \cap \delta$  is club in  $\delta$ . These assertions use only bounded quantification over the decoded sets and ordinals and are absolute to  $N$ .  $\square$

## 5.5 The complete local formula

Put  $\Delta(a) = \{2k + 1 + a(k) : k < \omega\}$ , using the characteristic function of  $a$ . If the header of  $a$  is valid, let  $\gamma = \gamma_a$ . In a decoding tuple of height  $\delta$ , and for  $m \in \Delta(a)$ , form inside  $N$

$$Z_m^{N,a} = \{\xi < \delta : |a \cap B_{\gamma+\xi, m}^P| < \omega\}, \quad X_m^{N,a} = \text{Odd}(Z_m^{N,a}).$$

Let  $X_{m,i}^{N,a} = \{\xi < \delta : 3\xi + i \in X_m^{N,a}\}$  for  $i < 3$ . The local test is the following first-order assertion in  $N$ : there is an ordinal  $\eta < \gamma$  with  $\omega\eta + \omega < \gamma$  such that for every  $m \in \Delta(a)$ ,

$$\begin{aligned} X_{m,1}^{N,a} &= \eta, & X_{m,2}^{N,a} &= m, \\ C_m^{N,a} &:= X_{m,0}^{N,a} \text{ is club in } \delta, & C_m^{N,a} \cap S_{\omega\eta+(m-1)}^P &= \emptyset. \end{aligned}$$

Here ordinals on the right are identified with their sets of predecessors. All existential objects in the test are elements of  $N$ , and all tests are arithmetic in a real code for the decoding tuple. The even-marker channel proves that the stage codes pass this test; no further clause about that channel is used in the formula.

**Definition 5.6.**  $\Theta_n(a)$  holds if  $a \cap 2\omega$  is infinite, its arithmetic header is a wellorder of type at least  $\omega$ , and every certified decoding tuple for  $a$  passes the preceding local test.

**Theorem 5.7** (Complexity and genuine ambient clubs). *The predicate  $\Theta_n$  is parameter-free  $\Pi_{n+2}^1$ . If it holds of  $a$ , there is a single ordinal  $\eta < \omega_1$  and, for every  $m \in \Delta(a)$ , a club  $C_m \subseteq \omega_1$  such that*

$$C_m \cap S_{\omega\eta+(m-1)} = \emptyset.$$

*All these objects are the coherent ambient decodings of  $a$ .*

*Proof.* Quantify universally over real codes for decoding tuples. Well-foundedness of the model code is  $\Pi_1^1$ , certification is  $\Pi_{n+1}^1$ , and the local assertion is arithmetic in the codes. The implication therefore has a  $\Sigma_{n+1}^1$  matrix, and the outer universal real quantifier gives  $\Pi_{n+2}^1$ . The additional  $\Pi_1^1$  header condition does not increase this upper bound.

For correctness, form the global sets  $Z_m, X_m, X_{m,i}$  by the same displayed formulas using the true array and all ordinals below  $\omega_1$ . Certified-array correctness implies that in every decoding tuple of height  $\delta$  the local sets are exactly the restrictions of these global sets to  $\delta$ . Every such height exceeds the fixed header ordinal  $\gamma$ . For any one selected column, the first side channel below  $\gamma$  therefore determines the same ordinal  $\eta < \gamma$  in every decoding tuple. The local test makes this ordinal common to all selected columns. Since decoding tuples exist at unbounded heights, the global side channels have exactly these prescribed values.

Set  $C_m = X_{m,0}$ . It is unbounded: given  $\beta < \omega_1$ , use a decoding tuple of height  $\delta > \beta$  and the unboundedness of  $C_m \cap \delta$ . It is closed: if  $\lambda < \omega_1$  is a limit point of  $C_m$ , choose such a tuple with  $\delta > \lambda$  and use the closure of  $C_m \cap \delta$ . Finally, if a point belonged to both  $C_m$  and the indicated stationary set, a tuple of sufficiently large height would see that point in both correctly decoded sets, contradicting the local test.  $\square$

**Theorem 5.8** (Genuine codes and persistence). *Suppose  $a$  has a valid header coding  $\gamma > \omega\eta + \omega$ , and for every  $m \in \Delta(a)$  it almost-disjointly codes on its tail column the set*

$$Y_m = \text{TailLoc}_\gamma(\text{Pack}(\eta, m, C_m)), \quad C_m \cap S_{\omega\eta+(m-1)} = \emptyset.$$

*Then  $\Theta_n(a)$  holds. It continues to hold in every later extension covered by theorem 3.9.*

*Proof.* In any certified decoding tuple, the header gives  $\gamma < \delta$ , and correct array decoding gives the actual tail of each  $Y_m$ . Lemma 5.5 puts  $\delta$  into that column's marker club, so its decoded objects are exactly  $\eta, m, C_m \cap \delta$ . They pass the local test. The same argument applies to every new decoding tuple in a later extension. Finiteness of intersections of fixed reals, the header's wellorder and its order type, and the closed unboundedness of the fixed clubs are unchanged by an  $\omega_1$ -preserving extension. Recognition and lemma 4.2 supply the same ground arrays. Thus every new tuple passes the test.  $\square$

## 6 A real-generated self-coding stage

The circularity is avoided by preparing both possible club codes for each bit in an auxiliary extension, and then passing to the complete Boolean subalgebra generated by the single mad real. The auxiliary extension is used to define the forcing; it is not the stage extension. The actual stage extension is  $V[a]$ . This distinction is essential: the auxiliary extension kills both stationary coordinates in each pair, whereas  $V[a]$  kills exactly the coordinate selected by  $a$ .

### 6.1 A bounded tail for all the hitting requirements

For any family  $\mathcal{H} \subseteq [\omega]^\omega$ , write  $\mathcal{I}(\mathcal{H})$  for the ideal generated by  $\mathcal{H}$  and Fin. Let

$$\mathcal{B}_{\geq \gamma} = \{B_{\xi, m} : \gamma \leq \xi < \omega_1, m < \omega\}.$$

Only pairwise almost disjointness of the array is needed in the next lemma. The family  $\mathcal{A}$  need not be countable for this argument.

**Lemma 6.1.** *Suppose that  $c_i \notin \mathcal{I}(\mathcal{A})$  for every  $i < \omega$ . There is  $\gamma < \omega_1$  such that, for every  $i < \omega$ , every finite  $D \subseteq \mathcal{A}$ , and every finite  $E \subseteq \mathcal{B}_{\geq \gamma}$ ,*

$$|c_i \setminus (\bigcup D \cup \bigcup E)| = \omega. \tag{T}$$

*Proof.* Fix  $i$ . If  $c_i \notin \mathcal{I}(\mathcal{A} \cup \mathcal{B})$ , no bound is needed for this  $i$ . Otherwise choose finite  $D_i \subseteq \mathcal{A}$  and  $E_i \subseteq \mathcal{B}$  such that

$$c_i \subseteq^* \bigcup D_i \cup \bigcup E_i.$$

The set  $E_i$  is nonempty, since  $c_i \notin \mathcal{I}(\mathcal{A})$ . Choose  $\gamma_i < \omega_1$  strictly above the first coordinates of all the indices of the members of  $E_i$ . Suppose toward a contradiction that finite  $D \subseteq \mathcal{A}$  and  $E \subseteq \mathcal{B}_{\geq \gamma_i}$  also satisfy  $c_i \subseteq^* \bigcup D \cup \bigcup E$ . Then

$$c_i \subseteq^* \bigcup (D_i \cup D) \cup \left( \bigcup E_i \cap \bigcup E \right).$$

The last intersection is finite: the two finite collections of array members have disjoint indices, and the entire array is almost disjoint. This would put  $c_i$  in  $\mathcal{I}(\mathcal{A})$ , a contradiction. Taking the supremum of these countably many bounds gives the required  $\gamma < \omega_1$ .  $\square$

The choice of a tail after the test sequence is known is the same positivity mechanism used in [4, Section 3]. The next construction also encodes that choice in the new real.

## 6.2 Inputs and independent localizers

Work in a model  $V$  of  $\text{ZFC} + \text{CH}$ . Let  $\mathcal{A}$  be the current countable almost disjoint family, and let  $\vec{c} = \langle c_i : i < \omega \rangle$  be positive modulo  $\mathcal{I}(\mathcal{A})$ . For an infinite real set  $B$ , let  $B(j)$  be its increasing enumeration and put  $E(B) = \{B(2j) : j < \omega\}$  and  $O(B) = \{B(2j+1) : j < \omega\}$ . These are enumeration halves, distinct from the ordinal channels used in the localizer. Assume the half-channel invariant

$$\forall D \in [\mathcal{A}]^{<\omega} \forall B \in \mathcal{B} \quad |E(B) \setminus \bigcup D| = |O(B) \setminus \bigcup D| = \omega. \quad (\text{H})$$

Fix a fresh block  $\eta$ . For  $m = 2k + 1 + \varepsilon \geq 1$ , abbreviate its stationary coordinates by  $S_m = S_{\iota(\eta, k, \varepsilon)}$ . Choose in  $V$  a countable limit ordinal  $\gamma > \omega\eta + \omega$  satisfying lemma 6.1. Fix in  $V$  a real  $z$  coding a wellorder of order type  $\gamma$ .

For each  $1 \leq m < \omega$ , let  $\mathbb{D}_m$  shoot a club  $C_m \subseteq \omega_1 \setminus S_m$ , and let

$$\mathbb{D} = \prod_{1 \leq m < \omega} \mathbb{D}_m$$

with countable support. Use the explicitly defined independent localizers of lemma 5.5: in  $V[C_m]$ , define

$$Y_m = \text{TailLoc}_\gamma(\text{Pack}(\eta, m, C_m)) \subseteq [\gamma, \omega_1). \quad (\text{L})$$

By lemma 5.5, the fixed decoding recovers  $(\eta, m, C_m)$  and has the required local correctness once the header has supplied  $\gamma$ . Crucially,  $Y_m$  belongs to  $V[C_m]$  and depends on no other club coordinate. It does not encode the as-yet-undefined real  $a$ .

By lemma 4.6,  $\mathbb{D}$  has size at most  $\omega_1$ , adds no countable sequences of ordinals, and is  $R$ -proper whenever  $R$  is stationary and  $R \cap S_m \in \text{NS}_{\omega_1}$  for every  $m$ . The same lemma applies to each subproduct.

## 6.3 The finite-stem forcing in the auxiliary extension

For  $t \in 2^{<\omega}$  let

$$s(t) = \{j < |t| : t(j) = 1\}, \quad \Delta(t) = \{2k + 1 + t(k) : k < |t|\}.$$

The map  $t \mapsto \Delta(t)$  is monotone under extension. Normalize the canonical array so that  $B_{0,0} = 2\omega$  and all other members lie on the odd integers. This is obtained by sending the other members of a canonical almost disjoint array to the odd integers and adjoining  $2\omega$ . The half-channel invariant then keeps both  $4\omega$  and  $4\omega + 2$  positive modulo every finite old subfamily. Let  $\text{rk}_{B_{0,0}}(2j) = j$ . A stem  $t$  satisfies the header rule if, writing  $h = s(t) \cap B_{0,0}$  in increasing order,

$$z(i) = 1 \iff h(2i) \in 4\omega \quad \text{whenever } 2i < |h|. \quad (\text{Z})$$

There is no restriction at the odd-indexed positions of  $h$ .

In  $V[\vec{C}]$ , define  $\mathbb{Q}$  as follows. A condition is  $q = (t_q, F_q, H_q)$ , where  $t_q$  is a finite binary stem satisfying (Z),  $F_q \in [\mathcal{A}]^{<\omega}$ , and

$$H_q \in [\{(\xi, m) : m \in \Delta(t_q), \xi \in Y_m\}]^{<\omega}.$$

Put  $q' \leq q$  if the stem and the two restraint coordinates extend their counterparts in  $q$ , and every new 1 in the stem avoids

$$\bigcup F_q \cup \bigcup_{(\xi, m) \in H_q} B_{\xi, m}. \quad (\text{A})$$

Let  $a$  be the union of the generic stems, identified with its set of 1's. The header set  $B_{0,0}$  is never itself a coding restraint, since every  $Y_m$  is supported at indices at least  $\gamma > 0$ .

**Lemma 6.2.** *The forcing  $\mathbb{Q}$  is  $\sigma$ -centered and has size at most  $\omega_1$ . It forces:*

- (1)  *$a$  is infinite, is almost disjoint from  $\mathcal{A}$ , and meets every  $c_i$  infinitely;*
- (2)  *$a \cap B_{0,0}$  is infinite, and  $(Z)$  recovers  $z$  arithmetically from  $a$  alone; the wellorder coded by  $z$  then determines  $\gamma$ ;*
- (3) *for each  $m \in \Delta(a)$  and every  $\xi < \omega_1$ ,*

$$\xi \in Y_m \iff |a \cap B_{\xi, m}| < \omega;$$

- (4) *the half-channel invariant (H) holds for  $\mathcal{A} \cup \{a\}$ .*

*Proof.* Conditions with the same stem have a common extension obtained by uniting their finite restraints. The cardinal bound follows because the stems are finite and all restraints are finite sets from an array of size  $\omega_1$  and the old family. Stems may always be extended by zeroes.

The following elementary extension procedure handles the header. If the next selected point of  $B_{0,0}$  would have even index, first add a sufficiently large point of the congruence class prescribed by (Z), outside the restraints (A). Such points exist by (H) and pairwise almost disjointness of the array. The next selected point of  $B_{0,0}$  then has odd index and is unrestricted. After this optional preliminary step, a sufficiently large point from any prescribed set positive modulo the current restraints may be put into the stem: if it lies in  $B_{0,0}$ , it occupies an unrestricted odd position; otherwise it does not affect the header.

For every  $i$  and every bound, adaptive-tail positivity gives a point of  $c_i$  outside (A) beyond the bound. Apply the procedure just described. It follows that  $a$  meets every  $c_i$  infinitely. It is dense to put any prescribed member of  $\mathcal{A}$  into  $F_q$ , which proves almost disjointness. Applying the header procedure repeatedly also shows that  $a \cap B_{0,0}$  is infinite and codes  $z$ .

Once a stem selects  $m$ , it is dense to put  $(\xi, m)$  in  $H_q$  for any  $\xi \in Y_m$ . This gives the finite-intersection direction. For  $\xi \notin Y_m$ , the set  $B_{\xi, m}$  retains infinitely many points outside (A): (H) handles the old-family restraints, and the other finitely many array restraints remove only finitely many points. The header procedure therefore gives arbitrarily large points of  $a \cap B_{\xi, m}$ . The argument is applied below any stem already selecting  $m$ , as required for the assertion about  $m \in \Delta(a)$ .

Finally fix a finite  $D \subseteq \mathcal{A}$ , an array member  $B$ , one of its halves  $K$ , and a bound. Choose a point of  $K \setminus \bigcup D$  beyond both the bound and the current stem, and extend the stem by zeroes through that point. This forces a fresh point outside  $a$ . These dense requirements prove (H) after adjoining  $a$ .  $\square$



## 6.4 The actual forcing

Let  $\mathbb{P}^{\text{aux}} = \mathbb{D} * \dot{\mathbb{Q}}$ , and let  $\mathbb{B}$  be its regular open completion. Let  $\mathbb{B}_a$  be the smallest complete Boolean subalgebra of  $\mathbb{B}$  containing

$$\|k \in \dot{a}\|_{\mathbb{B}} \quad (k < \omega).$$

For  $b \in \mathbb{B}$ , let

$$\pi(b) = \bigwedge \{d \in \mathbb{B}_a : b \leq d\}$$

be the canonical Boolean projection. Choose a dense set  $P_0 \subseteq \mathbb{P}^{\text{aux}}$  of size at most  $\omega_1$  and define

$$\mathbb{Q}^{\text{sc}} = \{\pi(p) : p \in P_0\} \setminus \{0\},$$

with the order inherited from  $\mathbb{B}_a$ . This is a dense suborder of  $\mathbb{B}_a$  of size at most  $\omega_1$ . To see that  $P_0$  exists, note that  $|\mathbb{D}| = \omega_1$  under CH. All candidate second-step conditions are finite tuples from a fixed ground-model set of size  $\omega_1$ . The pairs  $(d, \check{q})$  for which  $d$  forces that  $q$  is valid are dense. There are at most  $\omega_1$  such pairs. This counts the chosen dense forcing, not the whole Boolean completion. By the intermediate model theorem, complete generation by the digit values gives  $V[G \cap \mathbb{B}_a] = V[a]$ . In particular the actual stage retains precisely this intermediate extension. The auxiliary generic for an unused club coordinate is not retained.

**Lemma 6.3.** *Relative properness is inherited by complete Boolean subalgebras. Moreover, for  $p \in \mathbb{B} \setminus \{0\}$  the map*

$$d \mapsto d \wedge p$$

*is a complete embedding of  $\mathbb{B}_a \restriction \pi(p)$  into  $\mathbb{B} \restriction p$ . If the name  $\dot{a}$  on this cone belongs to a complete subalgebra  $\mathbb{C}$  of  $\mathbb{B} \restriction p$ , the image of the displayed embedding is contained in  $\mathbb{C}$ .*

*Proof.* For the first assertion, take a master condition in the larger algebra and project it to the smaller one. Every dense set of the smaller algebra lifts to a predense set of the larger algebra; Boolean projection therefore transfers the master-condition property. The countable models and their prescribed stationary heights are unchanged.

The displayed map preserves all joins, meets, and relative complements. It is injective because every nonzero  $d \leq \pi(p)$  meets  $p$ : otherwise  $p \leq \neg d$ , and the definition of  $\pi(p)$  would imply  $\pi(p) \leq \neg d$ . For the last assertion, the inverse image of  $\mathbb{C}$  under the restriction map  $b \mapsto b \wedge p$  is a complete subalgebra of  $\mathbb{B}$  containing all Boolean values of the bits of  $\dot{a}$ . By minimality it contains  $\mathbb{B}_a$ .  $\square$

## 6.5 Omitting an unselected coordinate

**Lemma 6.4.** *Fix  $j = 2k + 1 + \varepsilon \geq 1$  and put*

$$e_j = \|j \notin \Delta(a)\| = \|a(k) = 1 - \varepsilon\| \in \mathbb{B}_a.$$

*If  $e_j \neq 0$ , the algebra  $\mathbb{B}_a \restriction e_j$  completely embeds into an  $S_j$ -proper omission forcing.*

*Proof.* Let  $\mathbb{Q}^{-j}$  consist of those second-step conditions whose stem has length greater than  $k$  and has  $t(k) = 1 - \varepsilon$ . This is a dense presentation of the part of the second-step forcing below  $e_j$ . Every such stem and all its extensions omit  $j$  from  $\Delta$ . Thus  $\mathbb{Q}^{-j}$  never uses  $Y_j$ .

All other  $Y_m$  are defined in their individual club extensions. The poset  $\mathbb{Q}^{-j}$  is literally the same before and after forcing with  $\mathbb{D}_j$ : its conditions consist only of finite stems, finite sets from the old family, and finite ordinal restraints from the other  $Y_m$ . Therefore

$$(\mathbb{D} * \dot{\mathbb{Q}}) \restriction e_j \simeq (\mathbb{D}_{-j} * \dot{\mathbb{Q}}^{-j}) \times \mathbb{D}_j,$$

where  $\mathbb{D}_{-j}$  omits coordinate  $j$  and the last factor is the ground-model club-shooting poset. The real  $a$  is the union of the finite stems in the first factor; all its restricted bit-values therefore belong to that factor's complete Boolean algebra. By lemma 6.3, so does  $\mathbb{B}_a \restriction e_j$ .

The subproduct  $\mathbb{D}_{-j}$  is  $S_j$ -proper because  $S_j$  has nonstationary intersection with every remaining target. The poset  $\mathbb{Q}^{-j}$  is still  $\sigma$ -centered: conditions sharing a stem have a common extension by uniting their restraints. Its two-step iteration with  $\mathbb{D}_{-j}$  is therefore  $S_j$ -proper. Apply complete-subalgebra transfer once more.  $\square$

**Theorem 6.5** (Self-coding stage). *Use the independent localizers constructed in lemma 5.5. The forcing  $\mathbb{Q}^{\text{sc}}$  defined above has size at most  $\omega_1$  and has the following properties.*

- (1) *Its extension is  $V[a]$ , and  $a$  satisfies all conclusions of lemma 6.2.*
- (2) *It is  $S_*$ -proper and is  $R$ -proper for every old stationary  $R$  having nonstationary intersection with every coordinate of the fresh block. In particular it preserves every stationary coding coordinate outside this block.*
- (3) *Below the Boolean value of  $a(k) = \varepsilon$ , it is  $S_{2k+2-\varepsilon}$ -proper.*
- (4) *In its actual extension, for every  $k < \omega$  and  $\varepsilon < 2$ ,*

$$S_{2k+1+\varepsilon} \text{ is nonstationary} \iff \varepsilon = a(k).$$

- (5) *For each  $m \in \Delta(a)$ , the real  $a$  decodes the genuine localizer  $Y_m$  and therefore the club  $C_m$  and the common block index  $\eta$ . In every appropriate countable decoding model the header ensures  $\gamma < \omega_1^N$ , so the corresponding local decoding conclusions follow from lemma 5.5. If  $V[a]$  is an extension covered by theorem 3.9, then  $\Theta_n(a)$  holds, with persistence as in theorem 5.8.*

*Proof.* The size and real-generation assertions have already been proved. The finite-stem conclusions concern fixed real sets and finiteness of their intersections. They therefore transfer from the auxiliary extension to  $V[a]$ ; in particular the decoded  $Y_m$  is already in  $V[a]$  whenever  $m \in \Delta(a)$ .

The whole auxiliary forcing is  $R$ -proper for the stationary sets in clause (2), because its first step is the corresponding club product and its second step is proper. Apply lemma 6.3.

For clause (3), apply lemma 6.4 to  $j = 2k + 2 - \varepsilon$ . Its unselected-bit Boolean value is exactly  $\|a(k) = \varepsilon\|$ .

If  $m = 2k + 1 + a(k)$ , the real  $a$  decodes  $Y_m$ , hence  $C_m$ , and  $C_m \in V[a]$ . Closedness, unboundedness, and disjointness from  $S_m$  are absolute between these models with their common  $\omega_1$ . Thus  $S_m$  is nonstationary in  $V[a]$ . The other member of the pair remains stationary by clause (3), proving (4).

Finally a transitive decoding model containing  $a$  recovers  $z$  by the parameter-free arithmetic header rule, since  $B_{0,0} = 2\omega$ . A real in such a model cannot code a wellorder of type at least its own  $\omega_1$ . Hence  $\gamma < \omega_1^N$ . The localizer interface then supplies the asserted correct restrictions of the selected clubs and their common block index. The complete local formula and its persistence follow from theorem 5.8.  $\square$

The independence needed in the omission proof is established by the explicit localizer: its marker club is defined from  $C_m$  alone and the marker wellorder codes are fixed ground parameters. No other auxiliary coordinate enters  $Y_m$ . The full stage theorem therefore has no additional synchronization or localization hypothesis.

*Remark 6.6* (What has changed from the original condition scheme). The actual order is the Boolean order on the explicitly specified projected dense set  $\mathbb{Q}^{\text{sc}}$ . The finite stems occur in the auxiliary presentation used to prove its properties. Replacing this complete subforcing by the full auxiliary iteration would kill both members of every pair and would lose the exact block invariant. The subforcing is therefore part of the definition, not an optional factorization of the original proposed order.

## 6.6 No accidental self-codes

A block is *used* if it is assigned to a nontrivial self-coding stage. For a used block  $\eta$ , let  $a_\eta$  be the real added at that stage.

**Lemma 6.7.** *In the final model of the iteration from section 8:*

- (1) *if  $\eta$  is unused, every stationary set in  $\mathcal{S}^\eta$  is stationary;*
- (2) *if  $\eta$  is used, then for every  $k < \omega$  and  $\varepsilon < 2$ ,*

$$S_{i(\eta, k, \varepsilon)} \text{ is nonstationary} \iff \varepsilon = a_\eta(k).$$

*Proof.* By theorem 8.5, precisely the selected coordinates become nonstationary. Each block is used at most once.  $\square$

**Lemma 6.8.** *In the final model of the iteration,  $\Theta_n(a)$  holds only if  $a$  is the real assigned to a unique used block.*

*Proof.* Recognition holds by corollary 8.6. By theorem 5.7,  $a$  determines a block  $\eta$  and genuine clubs killing  $S_{\omega_\eta + (m-1)}$  for every  $m \in \Delta(a)$ . An unused block has no killed coordinate, so  $\eta$  is used. If its stage real is  $a_\eta$ , exactness of the block gives  $\Delta(a) \subseteq \Delta(a_\eta)$ . Both sets choose exactly one member of each pair  $\{2k+1, 2k+2\}$ , and therefore  $a = a_\eta$ . Reals added at different nontrivial stages are distinct: the later real is infinite and almost disjoint from the earlier one. This proves uniqueness.  $\square$

This is the final self-coding argument. The real chooses the columns from which it decodes clubs, and the exact block pattern recovers the same real without an external existential real witness.

## 7 Relative Mathias forcing

We give the Fin-case of the arguments in [1, Section 3]. Work in a model  $V$  of ZFC. Let  $T$  be a tree on  $2 \times \lambda$ , for an ordinal  $\lambda$ , and suppose that  $p[T] \subseteq [\omega]^\omega$  is an infinite almost disjoint family. Let  $\mathcal{I}(T)$  be the ideal generated by  $\text{Fin} \cup p[T]$ :

$$X \in \mathcal{I}(T) \iff \exists y_0, \dots, y_{m-1} \in p[T] \ X \subseteq^* \bigcup_{i < m} y_i.$$

**Definition 7.1.** The forcing  $\mathbb{M}_{\mathcal{I}(T)}$  consists of pairs  $(s, A)$  such that  $s \in [\omega]^{<\omega}$ ,  $A \in \mathcal{I}(T)^+$ , and  $\max(s) < \min(A)$ , with  $\max(\emptyset) = -1$ . Put  $(t, B) \leq (s, A)$  if

$$s \subseteq t \subseteq s \cup A, \quad B \subseteq A, \quad t \setminus s \subseteq A.$$

The generic real is the union of the finite stems.

**Lemma 7.2.** *If  $A_0 \supseteq A_1 \supseteq \dots$  are in  $\mathcal{I}(T)^+$ , there is  $A_\infty \in \mathcal{I}(T)^+$  such that  $A_\infty \subseteq^* A_k$  for every  $k < \omega$ .*

*Proof.* Write  $\mathcal{F} = \text{p}[T]$  and  $\mathcal{I} = \mathcal{I}(T)$ . We construct infinite sets  $B_j$  and distinct  $F_j \in \mathcal{F}$ . Suppose  $F_i$  has been chosen for  $i < j$ . Since every  $A_{j+k} \setminus \bigcup_{i < j} F_i$  is positive, choose increasing integers

$$b_k \in A_{j+k} \setminus \bigcup_{i < j} F_i \quad (k < \omega).$$

Put  $B_j = \{b_k : k < \omega\}$ . Then  $B_j \subseteq A_j$  and  $B_j \subseteq^* A_k$  for every  $k$ . If  $B_j \in \mathcal{I}^+$ , it is the required set. Otherwise, a finite subfamily of  $\mathcal{F}$  almost covers  $B_j$ , so there is  $F_j \in \mathcal{F}$  with  $|B_j \cap F_j| = \omega$ . It differs from the previously chosen members because  $B_j$  avoids their union.

If the construction does not stop, put

$$A_\infty = \bigcup_{j < \omega} (B_j \cap F_j).$$

For fixed  $k$ , the terms with  $j \geq k$  lie in  $A_k$ , and each of the finitely many earlier terms is almost contained in  $A_k$ . Thus  $A_\infty \subseteq^* A_k$ . Given a finite  $D \subseteq \mathcal{F}$ , choose  $F_j \notin D$ . Almost disjointness implies that infinitely many points of  $B_j \cap F_j$  lie outside  $\bigcup D$ . Hence  $A_\infty \in \mathcal{I}^+$ .  $\square$

**Lemma 7.3.** *The forcing  $\mathbb{M}_{\mathcal{I}(T)}$  is proper. Under CH it has size at most  $\omega_1$ .*

*Proof.* Put  $\mathcal{I} = \mathcal{I}(T)$  and  $\mathbb{U} = (\mathcal{I}^+, \subseteq^*)$ . Given a  $\subseteq^*$ -decreasing sequence, replace it by its finite intersections and apply lemma 7.2. Thus  $\mathbb{U}$  is countably closed and adds no reals.

Let  $H \subseteq \mathbb{U}$  be generic and put

$$\mathcal{U}_H = \{X \subseteq \omega : \exists A \in H (A \subseteq^* X)\}.$$

For each  $X \subseteq \omega$ , the conditions almost contained in  $X$  or in  $\omega \setminus X$  are dense in  $\mathbb{U}$ . It follows that  $\mathcal{U}_H$  is an ultrafilter. Every member of  $\mathcal{U}_H$  is  $\mathcal{I}$ -positive, and there are no new subsets of  $\omega$  at this step. In  $V[H]$ , let  $\mathbb{M}(\mathcal{U}_H)$  be Mathias forcing with reservoirs in  $\mathcal{U}_H$ . Conditions with the same stem have a common extension obtained by intersecting their reservoirs. Hence  $\mathbb{M}(\mathcal{U}_H)$  is  $\sigma$ -centered.

The map

$$(s, A) \mapsto (A, (\check{s}, \check{A})) \tag{M}$$

is a dense embedding of  $\mathbb{M}_{\mathcal{I}}$  into  $\mathbb{U} * \mathbb{M}(\dot{\mathcal{U}})$ , after taking separative quotients. To see density, strengthen a two-step condition to decide its second-step stem  $s$  and reservoir  $B$ . This is possible because  $\mathbb{U}$  adds no reals. If its first coordinate  $A$  forces  $B \in \dot{\mathcal{U}}$ , then  $A \cap B \in \mathcal{I}^+$ . Otherwise  $A \setminus B$  would be a positive extension forcing  $B \notin \dot{\mathcal{U}}$ . With

$$C = (A \cap B) \setminus (\max(s) + 1),$$

the image of  $(s, C)$  extends the given condition.

The map preserves order. If two images are compatible, a common extension has a stem extending both stems according to the Mathias order, and a reservoir in  $\mathcal{U}_H$  contained in both original reservoirs. Their intersection is therefore  $\mathcal{I}$ -positive. Removing its initial segment below that stem gives a common extension of the original conditions. Thus the map preserves incompatibility.

The two-step forcing is countably closed followed by ccc, and is therefore proper. Finally, a condition of  $\mathbb{M}_{\mathcal{I}}$  is a finite stem and a real reservoir, so  $|\mathbb{M}_{\mathcal{I}}| \leq 2^{\aleph_0}$ .  $\square$

**Lemma 7.4.** *Let  $G$  be  $\mathbb{M}_{\mathcal{I}(T)}$ -generic over  $V$ , and let  $x$  be its real. Then  $x$  is infinite and almost disjoint from every member of  $\mathcal{I}(T)^V$ . Every real definable in  $V[G]$  from  $[x]_{E_0}$  and ground-model parameters belongs to  $V$ .*

*Proof.* The sets of conditions with stems of size at least  $k$  are dense. For  $Y \in \mathcal{I}(T)^V$ , removing  $Y$  from a reservoir leaves a positive set, so it is dense to require that the reservoir avoid  $Y$ . This proves the first assertion.

For the last assertion, observe that, for stems  $s, t$  below the minimum of the same reservoir  $A$ , the map

$$(s \cup u, B) \mapsto (t \cup u, B)$$

is an isomorphism between the cones below  $(s, A)$  and  $(t, A)$ . It changes the generic real only on a finite set. Thus these two conditions force the same statements about  $[\dot{x}]_{E_0}$  and ground-model parameters.

Work below a condition  $(s, A_0)$  forcing that a fixed formula defines a real  $y$  from these parameters. Recursively choose  $A_{k+1} \subseteq A_k$  in  $\mathcal{I}(T)^+$  and  $\varepsilon_k < 2$  such that some extension  $(t, A_{k+1}) \leq (s, A_k)$  decides the  $k$ -th bit as  $\varepsilon_k$ . The isomorphism shows that  $(s, A_{k+1})$  makes the same decision. By lemma 7.2, choose  $A_\infty \in \mathcal{I}(T)^+$  with  $A_\infty \subseteq^* A_k$  for all  $k$ ; intersect with  $A_0$  and remove an initial segment if necessary.

We claim that  $(s, A_\infty)$  forces  $y(k) = \varepsilon_k$  for every  $k$ . Otherwise some  $(t, B) \leq (s, A_\infty)$  forces the opposite at a coordinate  $k$ . Remove finitely many points from  $B$  to put it inside  $A_{k+1}$ . Changing its stem back to  $s$  preserves that decision, contradicting the choice of  $A_{k+1}$ . Thus below  $(s, A_\infty)$ ,  $y$  equals the ground-model real  $\langle \varepsilon_k : k < \omega \rangle$ . Such conditions are dense, proving the assertion.  $\square$

For  $x \subseteq \omega$ , put

$$T^x = \{t \in T : \exists y \in p[T_{[t]}] \mid |x \cap y| = \omega\},$$

where  $T_{[t]}$  consists of the nodes compatible with  $t$ . It is easy to see that  $T^x$  is a pruned subtree of  $T$  and depends only on  $[x]_{E_0}$ . If  $t \in T^x$ , its witnessing branch belongs to  $[T^x]$ . In particular,

$$t \in T^x \implies \exists y \in p[(T^x)_{[t]}] \mid |x \cap y| = \omega. \quad (1)$$

**Lemma 7.5.**

$$\Vdash_{\mathbb{M}_{\mathcal{I}(T)}} |p[T^{\dot{x}}]| \leq 1.$$

*Proof.* Work in a generic extension  $V[G]$ , and suppose that  $p[T^x]$  contains two distinct reals. Choose nodes  $s_0, s_1 \in T^x$  whose binary parts are incompatible. There are extensions  $t_i \supseteq s_i$  and  $l < \omega$  such that

$$y_0 \cap y_1 \subseteq l \quad \text{whenever } y_i \in p[(T^x)_{[t_i]}] \quad (i < 2). \quad (2)$$

Indeed, if there were no such extensions, we could extend both nodes recursively, increase their lengths, and put a new integer into the intersection of their binary parts at each step. Their unions would be two branches of  $T$  with distinct projections and infinite intersection. This contradicts lemma 3.8.

Put

$$Y_i = \bigcup p[(T^x)_{[t_i]}] \quad (i < 2).$$

Each  $t_i$  is a ground-model node, and  $Y_i$  is definable from  $t_i, T$ , and  $[x]_{E_0}$ . Hence  $Y_i \in V$  by lemma 7.4. By (1),  $|x \cap Y_i| = \omega$ , while (2) gives  $Y_0 \cap Y_1 \subseteq l$ .

For every old set  $Y \subseteq \omega$ , the conditions whose reservoir is contained in  $Y$  or in  $\omega \setminus Y$  are dense: one of the two pieces of any positive reservoir is positive. Thus  $x \subseteq^* Y_0$ , since  $x \cap Y_0$  is infinite. It follows that  $x \cap Y_1$  is finite, a contradiction.  $\square$

**Theorem 7.6** (Mathias destruction). *The forcing  $\mathbb{M}_{\mathcal{I}(T)}$  adds an infinite real  $x$  such that*

$$\forall y \in p[T] \quad |x \cap y| < \omega,$$

*where the projection is interpreted in the extension. This assertion holds in every further forcing extension.*

*Proof.* The real  $x$  is infinite by lemma 7.4. Suppose  $y \in p[T]$  and  $|x \cap y| = \omega$  in  $V[G]$ . A branch witnessing  $y \in p[T]$  is then a branch of  $T^x$ . By lemma 7.5,  $p[T^x] = \{y\}$ , so  $y$  is definable from  $T$  and  $[x]_{E_0}$ . By lemma 7.4,  $y \in V$ . Membership of an old real in  $p[T]$  is absolute, so  $y \in \mathcal{I}(T)^V$ . The first assertion of the same lemma now gives  $|x \cap y| < \omega$ , a contradiction.

For persistence, work in  $V[G]$  and form the tree of finite sequences  $\langle (t_i, k_i) : i < r \rangle$  such that the  $t_i$  are properly increasing nodes of  $T$ , the  $k_i$  are increasing integers in  $x$ , and the binary part of  $t_i$  has value 1 at  $k_i$ . A branch through this tree exists exactly when some  $y \in p[T]$  meets  $x$  infinitely. The tree is therefore well-founded in  $V[G]$ . Its ordinal rank function remains a rank function in every further extension, which proves persistence.  $\square$

**Corollary 7.7** (Projective destroyer). *Assume the size bound of proposition 3.5, and let  $T = T(e, r)$ . Suppose  $p[T]$  is an infinite almost disjoint family in an intermediate model. Then  $\mathbb{M}_{\mathcal{I}(T)}$  adds an infinite real almost disjoint from the full later interpretation of the  $\Sigma_{n+2}^1(r)$  family represented by  $T$ .*

*Proof.* Apply theorem 7.6 to the fixed tree  $T$ . Its projection continues to represent the formula by proposition 3.5.  $\square$

By lemma 7.3, destruction stages preserve all stationary subsets of  $\omega_1$ .

## 8 The countable-support iteration

Work in  $W = M_n$  and let  $\kappa = \omega_1^W$ . All density bounds in this section are computed in  $W$ . We define a countable-support iteration

$$\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha < \kappa \rangle.$$

Fix a partition  $\kappa = E_{\text{sc}} \cup E_{\text{kill}}$  into two cofinal sets. The bookkeeping names are chosen during the recursion, as described in section 8.6. Let  $V_\alpha = W[G_\alpha]$ , and let  $\mathcal{A}_\alpha$  consist of the reals added at nontrivial self-coding stages below  $\alpha$ .

### 8.1 Self-coding stages

At  $\alpha \in E_{\text{sc}}$ , the bookkeeping supplies a sequence  $\langle \dot{c}_i : i < \omega \rangle$  of names for infinite subsets of  $\omega$ . Work in  $V_\alpha$ . If some  $c_i$  is covered modulo finite by a finite subfamily of  $\mathcal{A}_\alpha$ , let  $\mathbb{Q}_\alpha$  be trivial. Otherwise choose the least unused block  $\eta_\alpha$  and put

$$\mathbb{Q}_\alpha = \mathbb{Q}^{\text{sc}}(\mathcal{A}_\alpha, \langle c_i : i < \omega \rangle, \eta_\alpha).$$

At a nontrivial stage write  $a_\alpha$  for the generic real.

### 8.2 Destruction stages

At  $\alpha \in E_{\text{kill}}$ , the bookkeeping supplies a formula index  $e$  and a name  $\dot{r}$  for a real. In  $V_\alpha$ , take the section  $T = T(e, r)$  of the tree fixed in proposition 3.5. If  $p[T]$  is an infinite almost disjoint family, put

$$\mathbb{Q}_\alpha = \mathbb{M}_{\mathcal{I}(T)}.$$

Otherwise use the trivial forcing.

### 8.3 Dense subsets

We first prove the density estimates used in the induction. We use separative presentations of the iterands and identify conditions which are equivalent under forcing. Put  $\mathbb{B}_\alpha = \text{RO}(\mathbb{P}_\alpha)$  and let  $i_{\alpha\beta} : \mathbb{B}_\alpha \rightarrow \mathbb{B}_\beta$  be the embeddings induced by the iteration. Write  $[p]_\alpha$  for the Boolean value of  $p \in \mathbb{P}_\alpha$ .

**Lemma 8.1.** *Suppose  $\mathbb{P}$  has a dense subset of size at most  $\kappa$  and*

$$\Vdash_{\mathbb{P}} \text{dens}(\dot{\mathbb{Q}}) \leq \kappa.$$

*Then  $\mathbb{P} * \dot{\mathbb{Q}}$  has a dense subset of size at most  $\kappa$ .*

*Proof.* Let  $D \subseteq \mathbb{P}$  be dense,  $|D| \leq \kappa$ . Choose names  $\langle \dot{q}_\xi : \xi < \kappa \rangle$  which enumerate a dense subset of  $\dot{\mathbb{Q}}$ , allowing repetitions. The set

$$\{(d, \dot{q}_\xi) : d \in D, \xi < \kappa\}$$

is dense. Indeed, given  $(p, \dot{q})$ , strengthen  $p$  to choose  $\xi$  with  $\dot{q}_\xi \leq \dot{q}$ , and then strengthen into  $D$ .  $\square$

**Lemma 8.2.** *Let  $\lambda$  be a limit of countable cofinality, and let  $\langle \alpha_m : m < \omega \rangle$  increase cofinally in  $\lambda$ . For  $p \in \mathbb{P}_\lambda$ ,*

$$[p]_\lambda = \bigwedge_{m < \omega} i_{\alpha_m \lambda}([p \restriction \alpha_m]_{\alpha_m}). \quad (3)$$

*Proof.* The left side is below every term on the right. Conversely, let  $r \in \mathbb{P}_\lambda$  have Boolean value below the right side. The Boolean projection to  $\mathbb{B}_{\alpha_m}$  sends  $[r]_\lambda$  to  $[r \restriction \alpha_m]_{\alpha_m}$ . Hence

$$[r \restriction \alpha_m]_{\alpha_m} \leq [p \restriction \alpha_m]_{\alpha_m}.$$

The iteration order is separative: a failure of an inequality first occurs at a successor coordinate, where separativity of the iterand allows an extension incompatible with the other condition. Thus  $r \restriction \alpha_m \leq p \restriction \alpha_m$  for every  $m$ . The definition of the iteration order gives  $r \leq p$ . Conditions are dense in the Boolean algebra, which proves the reverse inequality.  $\square$

**Lemma 8.3.** *Assume CH. Let  $\lambda < \omega_1$  be a limit ordinal, and suppose  $\mathbb{P}_\lambda$  is  $R$ -proper for a stationary  $R \subseteq \omega_1$ . If every  $\mathbb{P}_\beta$ ,  $\beta < \lambda$ , has a dense subset of size at most  $\omega_1$ , then so does  $\mathbb{P}_\lambda$ .*

*Proof.* Choose an increasing cofinal sequence  $\langle \alpha_m : m < \omega \rangle$  in  $\lambda$  and dense sets  $D_m \subseteq \mathbb{P}_{\alpha_m}$  of size at most  $\omega_1$ . For  $\vec{X} \in \prod_{m < \omega} [D_m]^{\leq \omega}$ , put

$$b(\vec{X}) = \bigwedge_{m < \omega} i_{\alpha_m \lambda} \left( \bigvee_{d \in X_m} [d]_{\alpha_m} \right). \quad (4)$$

There are at most  $((\omega_1)^{\aleph_0})^{\aleph_0} = \omega_1$  such values. We claim that the nonzero ones are dense in  $\mathbb{B}_\lambda$ .

Fix  $p \in \mathbb{P}_\lambda$ . Choose a countable  $N \prec H_\theta$  containing  $p$ , the iteration, and the two sequences above, with  $N \cap \omega_1 \in R$  and  $N$  in the club witnessing  $R$ -properness. Let  $q \leq p$  be  $(N, \mathbb{P}_\lambda)$ -generic, and put

$$X_m = \{d \in D_m \cap N : d \leq p \restriction \alpha_m\}.$$

In  $N$ , choose a maximal antichain  $A_m \subseteq D_m$  whose members are either below  $p \restriction \alpha_m$  or incompatible with it. Its image is a maximal antichain in  $\mathbb{P}_\lambda$ , so  $A_m \cap N$  is predense below  $q$ . Since  $q \leq p$ , only the members below  $p \restriction \alpha_m$  can meet  $q$ . They belong to  $X_m$ , and therefore

$$[q]_\lambda \leq i_{\alpha_m \lambda} \left( \bigvee_{d \in X_m} [d]_{\alpha_m} \right).$$

It follows that  $0 < [q]_\lambda \leq b(\vec{X})$ . On the other hand,  $\bigvee_{d \in X_m} [d]_{\alpha_m} \leq [p \restriction \alpha_m]_{\alpha_m}$ , so lemma 8.2 gives  $b(\vec{X}) \leq [p]_\lambda$ . This proves the claim. Choose one condition below each nonzero value in (4). These conditions form a dense subset of size at most  $\omega_1$ .  $\square$

The sets  $D_m$  consist of iteration conditions, including their coordinate names. Thus the preceding count includes the names, not only the supports. It gives no bound on the size of the Boolean completion, and no such bound will be needed.

**Lemma 8.4.** *Assume CH. If  $\mathbb{P}$  is  $R$ -proper for some stationary  $R \subseteq \omega_1$  and has a dense subset of size at most  $\omega_1$ , then  $\mathbb{P}$  preserves CH.*

*Proof.* Fix a dense  $D \subseteq \mathbb{P}$  of size at most  $\omega_1$ . Given  $p$  and a name  $\dot{x}$  for a real, choose a countable test model  $N$  containing  $D, p, \dot{x}$ , and an  $(N, \mathbb{P})$ -generic  $q \leq p$ . For each digit, choose in  $N$  a maximal antichain contained in  $D$  deciding that digit. Its intersection with  $N$  is countable and predense below  $q$ . Thus below  $q$ ,  $\dot{x}$  is read from a countable ground-model set of labelled decisions in  $D \times \omega \times 2$ . There are at most  $(\omega_1)^{\aleph_0} = \omega_1$  such sets. For every real name these master conditions are dense, so their associated names cover all reals. By lemma 4.4,  $\omega_1$  is preserved. Hence CH holds in the extension.  $\square$

## 8.4 Preservation

For  $\alpha \leq \kappa$ , let  $\dot{K}_\alpha$  name the indices selected before  $\alpha$ . In  $V_\alpha$ ,

$$K_\alpha = \bigcup_{\substack{\beta < \alpha \\ \beta \text{ a nontrivial self-coding stage}}} \{\iota(\eta_\beta, k, a_\beta(k)) : k < \omega\}. \quad (5)$$

These sets increase with  $\alpha$  and are unions at limits.

**Theorem 8.5** (Preservation). *The iteration is defined through  $\kappa$ . For every  $\alpha \leq \kappa$ :*

- (1)  $\mathbb{P}_\alpha$  is  $S_*$ -proper, has a dense subset of size at most  $\kappa$  in  $W$ , and preserves  $\omega_1$  and CH;
- (2)  $\mathbb{P}_\alpha$  forces that  $\mathcal{A}_\alpha$  is almost disjoint and satisfies the half-channel invariant (H);
- (3) if  $i < \kappa$  and  $p \in \mathbb{P}_\alpha$  satisfy  $p \Vdash i \notin \dot{K}_\alpha$ , then  $\mathbb{P}_\alpha \restriction p$  is  $S_i$ -proper;
- (4) for every  $i < \kappa$ ,

$$\Vdash_{\mathbb{P}_\alpha} (S_i \text{ is nonstationary} \longleftrightarrow i \in \dot{K}_\alpha). \quad (6)$$

*Proof.* We prove the assertions by induction, together with the construction. At zero they hold because  $W \models \text{CH}$ , the coding sets are stationary, and  $\mathcal{A}_0 = \emptyset$ .

Suppose first that  $\alpha = \beta + 1$ . In  $V_\beta$ , the family  $\mathcal{A}_\beta$  is countable and satisfies (H). Every coordinate of an unused block is stationary by (4) at  $\beta$ . Hence theorem 6.5 applies whenever the self-coding test is positive. It gives  $S_*$ -properness, a dense presentation of size at most  $\kappa$ , and preservation of almost disjointness and (H). At a destruction stage these conclusions follow from lemma 7.3; the family and its intersections with fixed array members do not change. The trivial case needs no argument. By lemma 4.4,  $\mathbb{P}_\alpha$  is  $S_*$ -proper. Apply lemmas 8.1 and 8.4, in that order, to obtain its density bound and CH. This proves (1) and (2) at  $\alpha$ .

Let now  $\alpha < \kappa$  be a limit. The relative iteration theorem first gives  $S_*$ -properness. Apply lemma 8.3 in  $W$ , and then lemma 8.4. Every finite subfamily of  $\mathcal{A}_\alpha$  is contained in some  $\mathcal{A}_\beta$ ,  $\beta < \alpha$ . Its intersections with fixed reals do not change, so almost disjointness and (H) pass to the limit.

At  $\alpha = \kappa$ , relative properness follows in the same way. Every condition has countable, hence bounded, support. If  $D_\beta \subseteq \mathbb{P}_\beta$  is dense of size at most  $\kappa$  for  $\beta < \kappa$ , then the embedded



union  $\bigcup_{\beta < \kappa} D_\beta$  is dense in  $\mathbb{P}_\kappa$  and has size at most  $\kappa$ . Apply lemma 8.4. The argument for (2) is the same as at countable limits.

It remains to prove (3) and (4) at  $\alpha > 0$ . Suppose

$$p \Vdash_{\mathbb{P}_\alpha} i \notin \dot{K}_\alpha.$$

For every  $\beta < \alpha$ ,

$$p \restriction (\beta + 1) \Vdash_{\mathbb{P}_{\beta+1}} i \notin \dot{K}_{\beta+1}. \quad (7)$$

Indeed, a condition below this restriction forcing the opposite could be combined with the tail of  $p$ , contradicting the choice of  $p$ . The same assertion holds for restrictions to  $\beta$ .

Represent  $\mathbb{P}_\alpha \restriction p$  by the iteration restricted along  $p$ . Its initial forcing at  $\beta$  is equivalent to  $\mathbb{P}_\beta \restriction (p \restriction \beta)$ , and its next iterand is  $\mathbb{Q}_\beta \restriction p(\beta)$ , with the top condition used off the support of  $p$ . By induction the initial forcing preserves  $S_i$ . Work in one of its generic extensions. If the next iterand is trivial or a destroyer, its restriction is proper, hence  $S_i$ -proper. If it is self-coding in a block not containing  $i$ , apply the outside-block clause of theorem 6.5.

In the remaining case, write  $i = \iota(\eta_\beta, k, \varepsilon)$ . By (7),  $p(\beta)$  forces  $a_\beta(k) \neq \varepsilon$ . Its cone lies below the unselected-bit Boolean value, so lemma 6.4 gives  $S_i$ -properness. Thus every iterand of the restricted iteration is forced to be  $S_i$ -proper. Apply lemma 4.4 to prove (3), also when  $\alpha$  is a limit.

If  $i \in K_\alpha$ , a club disjoint from  $S_i$  was added at the stage selecting  $i$ . It remains a club because all later forcings preserve  $\omega_1$ . If a condition forces  $i \notin \dot{K}_\alpha$ , its cone preserves  $S_i$  by (3). Conditions deciding membership in  $\dot{K}_\alpha$  are dense. This proves (4) and completes the induction.  $\square$

The assertion in (3) concerns a cone where  $i$  is not selected. The whole forcing need not be  $S_i$ -proper, since another cone may select  $i$ . The unused blocks and the unselected coordinates of used blocks are both covered by (3).

## 8.5 Recognition and decoding along the iteration

**Corollary 8.6.** *For every  $\alpha \leq \kappa$ , the recognized structures in  $V_\alpha$  are those of  $W$ , and their limit unions compute the same canonical arrays. Each section  $T(e, r)$ , once  $r$  appears, represents its  $\Sigma_{n+2}^1(r)$  definition in every later intermediate model and in  $V_\kappa$ . Every self-coding stage real satisfies  $\Theta_n$  in all later models of the iteration.*

*Proof.* Take  $\rho = (\kappa^+)^W < \delta_0$ . By theorem 8.5, for all  $\alpha \leq \kappa$ ,

$$\text{dens}^W(\mathbb{P}_\alpha) \leq \kappa < \rho, \quad \omega_1^{V_\alpha} = \kappa.$$

Apply theorem 3.9 and proposition 3.5 to each forcing, viewed over  $W$ . The assertions about limit unions and canonical computations follow from lemmas 4.2 and 5.2. The stage theorem and theorem 5.8 now give the last assertion.  $\square$

## 8.6 Bookkeeping

We require the schedule to satisfy the following two clauses.

**Requirement 8.7** (Mad bookkeeping). Every countable sequence of infinite subsets of  $\omega$  in the final model is presented at a self-coding stage after the sequence has appeared.

**Requirement 8.8** (Destruction bookkeeping). Every pair  $(e, r) \in \omega \times \mathbb{R}$  in the final model is presented cofinally often at destruction stages. If the final projection  $p[T(e, r)]$  is infinite, one presentation occurs after a countably infinite subset of that projection has appeared.

**Lemma 8.9.** *Every real and every countable sequence of reals in  $V_\kappa$  belongs to some  $V_\delta$ ,  $\delta < \kappa$ .*

*Proof.* Fix a  $\mathbb{P}_\kappa$ -name  $\dot{x}$  for a real and a condition  $p$ . Choose a countable  $N \prec H_\theta$  containing the iteration,  $p$ , and  $\dot{x}$ , with  $\delta = N \cap \kappa \in S_*$  and  $N$  in the club witnessing relative properness. By theorem 8.5, there is an  $(N, \mathbb{P}_\kappa)$ -generic  $q \leq p$ . For each digit  $k$ , choose in  $N$  a maximal antichain  $A_k$  deciding that digit. The set  $A_k \cap N$  is countable and predense below  $q$ .

If  $s \in \mathbb{P}_\kappa \cap N$ , then  $\text{supp}(s) \in N$  is countable, so  $\text{supp}(s) \subseteq \delta$ . Thus  $s$  belongs to the embedded copy of  $\mathbb{P}_\delta$ . The labelled sets  $A_k \cap N$  therefore define a  $\mathbb{P}_\delta$ -name  $\dot{y}$  with  $q \Vdash \dot{x} = \dot{y}$ . Such  $q$  are dense as  $p$  varies. This proves the assertion for reals. Code a countable sequence of reals by one real to obtain the second assertion.  $\square$

**Lemma 8.10.** *The schedule can be chosen to satisfy requirements 8.7 and 8.8.*

*Proof.* By theorem 8.5, each  $\mathbb{P}_\beta$  forces CH. Once  $\mathbb{P}_\beta$  has been defined, choose names  $\dot{F}_\beta^{\text{sc}}$  and  $\dot{F}_\beta^{\text{kill}}$  for surjections from  $\kappa$  onto, respectively, the sequences in  $([\omega]^\omega)^\omega$  and the pairs in  $\omega \times \mathbb{R}$  of  $V_\beta$ .

For each  $t \in \{\text{sc}, \text{kill}\}$ , fix a partition

$$E_t = \bigcup_{\beta, \xi < \kappa} E_{\beta, \xi}^t$$

into pairwise disjoint cofinal sets. At a stage  $\alpha \in E_{\beta, \xi}^t$  with  $\beta < \alpha$ , present the entry  $\dot{F}_\beta^t(\xi)$ . At a remaining stage present a fixed entry from the ground-model list of the corresponding type. Every entry is therefore repeated cofinally often after  $\beta$ . The choice of  $\dot{F}_\beta^t$  uses only the iteration through  $\beta$ , so this specifies the recursion.

By lemma 8.9, every final sequence and every final parameter occurs in a bounded intermediate model. Hence it is scheduled later. If a final projection is infinite, choose a sequence of distinct members and apply bounded appearance to that sequence together with its parameter. A later repeated presentation of the formula and parameter then gives the last clause of requirement 8.8.  $\square$

Let  $G \subseteq \mathbb{P}_\kappa$  be generic and put

$$V^* = W[G], \quad \mathcal{A}^* = \{a_\alpha : \alpha \in E_{\text{sc}} \text{ and stage } \alpha \text{ is nontrivial}\}.$$

## 9 Definability

**Lemma 9.1.** *Every member of  $\mathcal{A}^*$  satisfies  $\Theta_n$  in  $V^*$ .*

*Proof.* Apply corollary 8.6 to the stage at which the real was added.  $\square$

**Lemma 9.2.** *If  $V^* \models \Theta_n(a)$ , then  $a \in \mathcal{A}^*$ .*

*Proof.* Apply lemma 6.8, using the stationary pattern from theorem 8.5 and recognition from corollary 8.6.  $\square$

**Theorem 9.3** (Definability). *In  $V^*$ ,*

$$\mathcal{A}^* = \{a \in [\omega]^\omega : \Theta_n(a)\}.$$

*Hence  $\mathcal{A}^*$  is parameter-free  $\Pi_{n+2}^1$ .*

*Proof.* Combine theorem 5.7 and lemmas 9.1 and 9.2.  $\square$

## 10 Almost disjointness and $\omega$ -maximality

**Lemma 10.1.** *The family  $\mathcal{A}^*$  is almost disjoint.*

*Proof.* At each nontrivial self-coding stage the new real is infinite and almost disjoint from every member of the current family. Finiteness of intersections of fixed reals is preserved by further forcing.  $\square$

**Theorem 10.2** ( $\omega$ -maximality). *The family  $\mathcal{A}^*$  is infinite and  $\omega$ -mad.*

*Proof.* Let  $\langle c_i : i < \omega \rangle$  be a sequence in  $V^*$  such that no  $c_i$  is covered modulo finite by a finite subfamily of  $\mathcal{A}^*$ . By lemmas 8.9 and 8.10, the sequence is presented at a self-coding stage after it has appeared. No  $c_i$  is covered by the smaller family present at that stage, so the stage is nontrivial. Its real meets every  $c_i$  infinitely by theorem 6.5.

By (H), no finite subfamily of  $\mathcal{A}_\alpha$  covers  $\omega$  modulo finite. The constant sequence  $c_i = \omega$  is repeated cofinally often by lemma 8.10. Each such stage is nontrivial, and its real differs from all preceding stage reals by almost disjointness. Thus  $\mathcal{A}^*$  is infinite.  $\square$

## 11 No boldface $\Sigma_{n+2}^1$ mad family

**Theorem 11.1.** *There is no infinite boldface  $\Sigma_{n+2}^1$  mad family in  $V^*$ .*

*Proof.* Suppose

$$\mathcal{F} = \{x \in [\omega]^\omega : \varphi_e(x, r)\}$$

is an infinite  $\Sigma_{n+2}^1(r)$  almost disjoint family in  $V^*$ . Choose a sequence of distinct members of  $\mathcal{F}$ . By lemma 8.9, this sequence and  $r$  belong to some  $V_\alpha$ . The tree  $T = T(e, r)$  is then available in  $V_\alpha$ . By corollary 8.6, it represents  $\varphi_e(\cdot, r)$  in every later model, including  $V^*$ .

Membership of a fixed real in the projection of a fixed tree is absolute. Hence the chosen members already belong to  $p[T]^{V_\alpha}$ , which is therefore infinite. It is almost disjoint by lemma 3.8. The bookkeeping presents  $(e, r)$  at a later destruction stage. The relative Mathias real added there is infinite and almost disjoint from every member of the projection in all later extensions, by corollary 7.7. In particular, it is almost disjoint from every member of  $\mathcal{F}$ . Thus  $\mathcal{F}$  is not maximal.  $\square$

## 12 Conclusion

*Proof of theorem 1.1.* Let  $G \subseteq \mathbb{P}_\kappa$  be generic over  $M_n$ . By theorem 8.5,  $M_n[G]$  preserves  $\omega_1$  and satisfies CH. By theorems 9.3 and 10.2, its family  $\mathcal{A}^*$  is infinite,  $\omega$ -mad, and parameter-free  $\Pi_{n+2}^1$ . By theorem 11.1, there is no infinite boldface  $\Sigma_{n+2}^1$  mad family in this extension.  $\square$

## 13 Open problems

We finish with some questions suggested by the proof.

1. *What is the consistency strength of the theorem?* The present proof starts from  $M_n$ . It is not clear whether this is necessary. The tree argument used to destroy  $\Sigma_{n+2}^1$  mad families requires less: the finite homogeneity argument can be run from  $n - 1$  Woodin cardinals together with a measurable cardinal above them. The remaining use of  $M_n$  lies in the recognition of the lower part at complexity  $\Pi_{n+1}^1$ . It is therefore natural to ask whether the theorem can be obtained from  $n - 1$  Woodin cardinals and a measurable above them. More strongly, can it be forced over  $L$ , or from assumptions below this level?

2. *What is the corresponding lightface problem?* Can one force a parameter-free  $\Pi_{n+2}^1$  mad family while ruling out only lightface  $\Sigma_{n+2}^1$  mad families from weaker hypotheses? Since there are only countably many lightface requirements, this separates the exact placement problem from the additional difficulty caused by arbitrary real parameters.
3. *Are there lower bounds from exact placement?* The mice  $M_n$  calibrate the projective complexity of canonical wellorders of the reals. It would be interesting to know whether an analogous converse holds for maximal almost disjoint families. For example, does the existence of a  $\Pi_{n+2}^1$  mad family together with the nonexistence of boldface  $\Sigma_{n+2}^1$  mad families imply the existence of an inner model with some fixed amount of large-cardinal strength?
4. *How does projective uniformization constrain the possible placement of mad families?* Törnquist's argument lifts in the presence of the corresponding uniformization theorem: if  $\Pi_m^1$  uniformization holds, then the existence of a  $\Sigma_{m+1}^1(a)$  mad family implies the existence of a  $\Pi_m^1(a)$  mad family. Thus some apparent placement patterns are ruled out by uniformization. One may ask for a classification of the possible first projective levels of mad families under specified uniformization principles, and in particular what can happen when these principles fail.
5. *Can exact placement be combined with a large continuum or prescribed cardinal characteristics?* The present construction preserves CH. The methods of Fischer–Friedman–Zdomskyy show that projective mad families can coexist with large continuum. Can the exact separation proved here be combined with, for example,  $2^{\aleph_0} = \aleph_2$ , or with prescribed values of  $\mathfrak{b}$ ,  $\mathfrak{d}$ , or related characteristics?
6. *Does the same phenomenon occur for other maximal objects?* The proof combines lower-part recognition, self-coding by stationary sets, and destruction of the competing pointclass by a forcing adapted to a fixed tree representation. It is natural to ask for analogous exact placement results for other definable maximal families, for example maximal orthogonal families, maximal independent families, or maximal cofinitary groups.

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